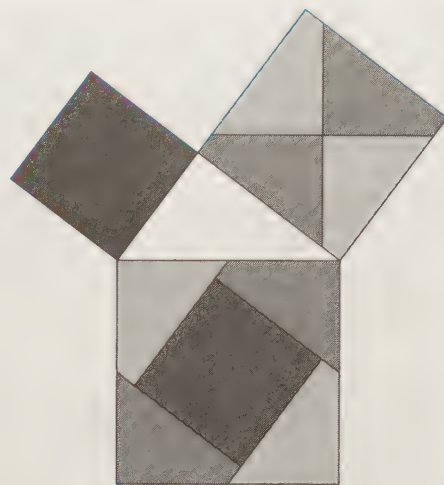




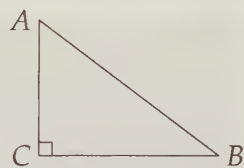
The Pythagorean Theorem

*Choose always the way that seems the best however rough it may be; custom will soon render it easy and agreeable.*  
– Pythagoras

# CHAPTER 6

## Right Triangles

In this chapter we study right triangles. As a quick refresher, a right triangle is a triangle that has a right angle among its angles. The side opposite the right angle is called the **hypotenuse**, and the other two sides are called **legs**. In the figure to the right,  $\overline{AB}$  is the hypotenuse, while  $\overline{AC}$  and  $\overline{BC}$  are both legs. As you'll see throughout the book, many problems are solved by building right triangles and using the principles you'll learn in this chapter, particularly one of the most famous math theorems of all: the Pythagorean Theorem.

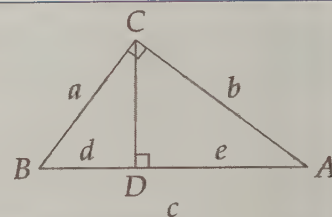


**Extra!** It has long been a sport among lovers of mathematics to find new proofs of the Pythagorean Theorem. Most engaging are the proofs that can be expressed simply as a diagram, challenging the viewer to fill in the details and learn how the diagram provides a proof of the Pythagorean Theorem. Pure mathematicians are not the only people who get to join in this game; proofs have been attributed to President James A. Garfield and Leonardo da Vinci (among many others). We'll share several of these 'proofs without words' with you in this book. You can find a long list of Pythagorean Theorem proofs at Alexander Bogomolny's excellent Interactive Mathematics Miscellany website and in Roger Nelsen's two *Proofs Without Words* books.

# 6.1 Pythagorean Theorem

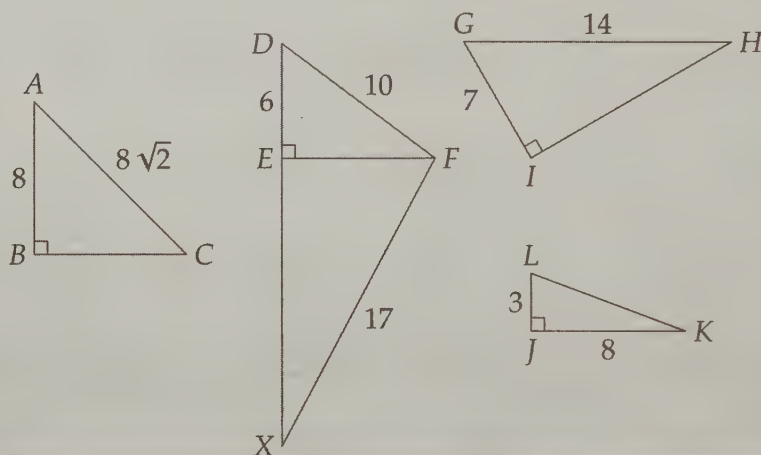
## Problems

**Problem 6.1:** In this problem we will prove one of the most famous theorems of mathematics. The diagram shows right triangle  $\triangle ABC$  with hypotenuse  $AB = c$  and legs  $AC = b$  and  $BC = a$ .

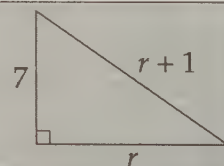


- Prove that  $a^2 = cd$ .
- Prove that  $b^2 = ce$ .
- Use the first two parts to show that  $a^2 + b^2 = c^2$ . This is the **Pythagorean Theorem**.

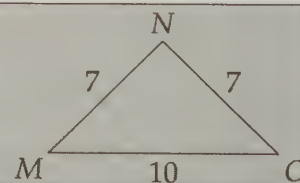
**Problem 6.2:** Find the missing sides in the triangles below.



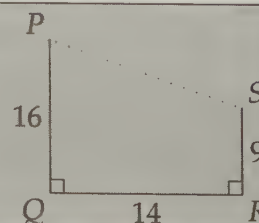
**Problem 6.3:** Find  $r$ .



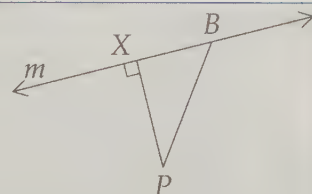
**Problem 6.4:** Isosceles triangle  $\triangle MNO$  has  $MN = NO = 7$  and  $MO = 10$ . Find  $[MNO]$ .



**Problem 6.5:** Two vertical poles are 14 feet apart. One is 16 feet tall, and the other is 9 feet tall. A rope extends straight from the top of one pole to the top of the other pole. How long is the rope?



**Problem 6.6:** Let  $P$  be a point and  $m$  be a line that doesn't pass through  $P$ . Let  $X$  be on  $m$  such that  $\overline{PX} \perp m$ . The distance from a point to a line is the length of the shortest segment from the point to the line. In this problem, we prove that the distance from  $P$  to  $m$  is  $PX$ .

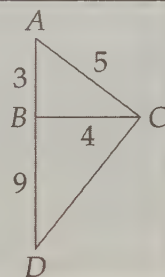


- Let  $B$  be a point on  $m$  besides  $X$ . What kind of triangle is  $\triangle PXB$ ?
- Use the Pythagorean Theorem to get an expression for  $PB^2$ .
- Why does the previous part show that  $PX$  is the shortest possible distance from  $P$  to line  $m$ ?

**Problem 6.7:** In this problem we investigate whether  $\triangle ABC$  must be a right triangle given that  $BC = a$ ,  $AC = b$ ,  $AB = c$ , and  $a^2 + b^2 = c^2$ .

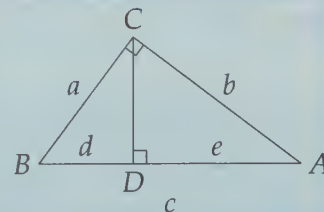
- Suppose we have a triangle  $\triangle DEF$  such that  $\angle F = 90^\circ$ ,  $EF = a$ , and  $DF = b$ . (These are the same lengths as  $BC$  and  $AC$  in  $\triangle ABC$ .) What is  $DE$ ?
- What can we say about triangles  $\triangle ABC$  and  $\triangle DEF$ ?
- Must  $\triangle ABC$  be a right triangle?

**Problem 6.8:** The side lengths are as marked in the diagram. Find  $CD$ .



We start off with a proof of one of the most famous theorems in mathematics.

**Problem 6.1:** Right triangle  $\triangle ABC$  has hypotenuse  $AB = c$  and legs  $AC = b$  and  $BC = a$ . Use similar triangles formed by altitude  $\overline{CD}$  to prove that  $a^2 + b^2 = c^2$ .



*Solution for Problem 6.1:* As we showed in Problem 5.14, we can use  $\triangle ADC \sim \triangle ACB$  to show that  $b/e = c/b$ , so  $b^2 = ce$ . Similarly,  $\triangle BDC \sim \triangle BCA$  gives us  $a/d = c/a$ , so  $a^2 = cd$ . Adding these expressions for  $a^2$  and  $b^2$  gives us

$$a^2 + b^2 = cd + ce = c(d + e) = c^2.$$

□

**Extra!** *Mighty is geometry; joined with art, resistless.*

—Euripides

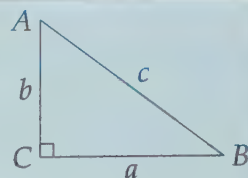
**Important:**



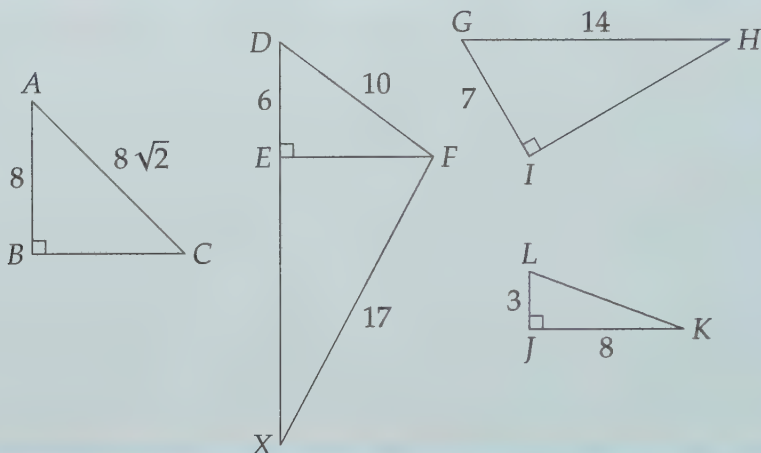
In any right triangle, the sum of the squares of the legs equals the square of the hypotenuse. In the figure to the right, we have

$$a^2 + b^2 = c^2$$

This is the famous **Pythagorean Theorem**.



**Problem 6.2:** Find the missing sides in the triangles below.

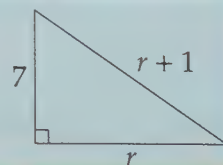


*Solution for Problem 6.2:*

- (a)  $\triangle ABC$ :  $AB^2 + BC^2 = AC^2$ , so  $64 + BC^2 = 128$ . Therefore,  $BC^2 = 64$  and  $BC = 8$ .
- (b)  $\triangle DEF$  &  $\triangle FEX$ : From  $\triangle DEF$  we have  $DE^2 + EF^2 = DF^2$ , so  $36 + EF^2 = 100$ , and  $EF = 8$ . Now we can tackle  $EX$  with  $\triangle FEX$ :  $EX^2 + FE^2 = FX^2$ , so  $EX^2 = FX^2 - EF^2 = 225$ . So,  $EX = 15$ .
- (c)  $\triangle GHI$ :  $HI^2 + GI^2 = GH^2$ , so  $HI^2 = 196 - 49 = 147$ . Therefore,  $HI = \sqrt{147} = 7\sqrt{3}$ .
- (d)  $\triangle JKL$ :  $KL^2 = LJ^2 + JK^2 = 9 + 64 = 73$ . So,  $KL = \sqrt{73}$ .

□

**Problem 6.3:** Find  $r$ .



*Solution for Problem 6.3:* We have a right triangle, so we can apply the Pythagorean Theorem:

$$7^2 + r^2 = (r + 1)^2.$$

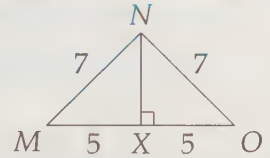
Therefore,  $49 + r^2 = r^2 + 2r + 1$ , so  $r = 24$ . □

Sometimes it's not the sides of a right triangle we are missing, but the whole right triangle itself! In these cases, we have to build the right triangle ourselves, then apply the Pythagorean Theorem.



**Problem 6.4:** Isosceles triangle  $\triangle MNO$  has  $MN = NO = 7$  and  $MO = 10$ . Find  $[MNO]$ .

*Solution for Problem 6.4:* Finding an altitude will allow us to find the area. When we first studied isosceles triangles on page 67, we saw that we can cut an isosceles triangle in half with an altitude. We do so here in the figure to the right, in which altitude  $\overline{NX}$  cuts  $\overline{MO}$  in half so that  $MX = XO = 5$ . The Pythagorean Theorem applied to  $\triangle MNX$  gives



$$MX^2 + NX^2 = MN^2,$$

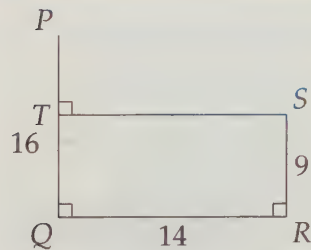
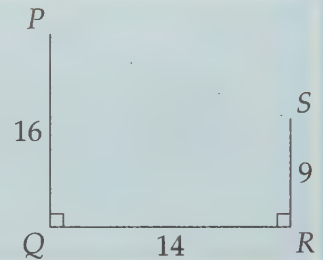
so  $25 + NX^2 = 49$ . Therefore,  $NX = \sqrt{24} = 2\sqrt{6}$ , and our area is

$$[MNO] = \frac{(NX)(MO)}{2} = 10\sqrt{6}.$$

□

In our next example, it's not as clear initially how to build a useful right triangle.

**Problem 6.5:** Two vertical poles are 14 feet apart. One is 16 feet tall, and the other is 9 feet tall. A rope extends straight from the top of one pole to the top of the other pole. How long is the rope?



*Solution for Problem 6.5:* We don't have a right triangle, but the right angles at  $Q$  and  $R$  have us thinking about building our own. We draw  $\overline{ST}$  perpendicular to  $\overline{PQ}$ , and we have a right triangle with hypotenuse  $\overline{PS}$ ! Since  $\angle Q + \angle STQ = 180^\circ$ , we have  $\overline{QR} \parallel \overline{TS}$ . Therefore,  $\angle TSR + \angle R = 180^\circ$ , so  $\angle TSR = 90^\circ$ , and  $STQR$  is a rectangle. Since  $STQR$  is a rectangle, we have  $TS = QR = 14$  and  $TQ = SR = 9$ . Therefore,  $PT = PQ - SR = 7$ , and we can now use the Pythagorean Theorem to find  $PS$ :

$$PS^2 = PT^2 + TS^2 = 49 + 196 = 245.$$

Therefore, the length of our rope is  $\sqrt{245} = 7\sqrt{5}$ . □

**Concept:** Building right triangles and applying the Pythagorean Theorem is one of the most common ways to find lengths. This is particularly true in problems that already involve right angles, such as Problem 6.5.

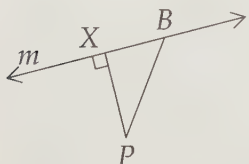
**Problem 6.6:** The distance from a point to a line is the length of the shortest segment from the point to the line. Prove that this shortest segment is perpendicular to the line.

*Solution for Problem 6.6:* This is a great example of proving the obvious. Sometimes the obvious is so obvious that people will propose solutions like this:

**Bogus Solution:** The shortest distance from  $P$  to the line  $m$  will be when we go straight from  $P$  to the line. This would make a right angle with  $m$  because we get a right angle when we go straight from the point to the line.



This ‘proof’ doesn’t say anything at all. There aren’t any false statements, but there aren’t any useful statements either. It essentially argues ‘the angle is a right angle because that’s what you get when you make a right angle.’ Here’s why we get a right angle:



We start by drawing a diagram. Let  $P$  be our point and  $m$  be our line. Let  $X$  be on  $m$  such that  $\overline{PX} \perp m$ . We must show that  $\overline{PX}$  is the shortest segment from  $P$  to  $m$ , so we must show that  $\overline{PX}$  is shorter than any other segment from  $P$  to  $m$ . We let  $B$  be another point on  $m$ , and compare  $PX$  to  $PB$ .

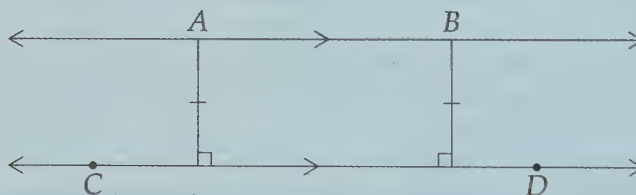
Now we have a right triangle,  $\triangle PXB$ , so we can use the Pythagorean Theorem:  $PB^2 = PX^2 + XB^2$ . Since  $XB^2$  is clearly greater than 0, we must have  $PB^2 > PX^2$ , so  $PB > PX$ . So, the perpendicular segment from  $P$  to  $m$  is shorter than any other segment from  $P$  to  $m$ .  $\square$

**Important:** The distance from a point to a line is the length of the segment from the point to the line that is perpendicular to the line.



You’ll have a chance to use this definition of the distance from a point to a line to prove the following as an Exercise:

**Important:** If  $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ , then  $A$  and  $B$  are the same distance from  $\overleftrightarrow{CD}$ .



Throughout this section we have used the Pythagorean Theorem to find information about the sides of right triangles. Perhaps you’re wondering if we can use the Pythagorean Theorem ‘in reverse’ to determine if a triangle is a right triangle.

**Problem 6.7:** Suppose we have  $\triangle ABC$  such that  $BC = a$ ,  $AC = b$ ,  $AB = c$ , and  $a^2 + b^2 = c^2$ . Is  $\triangle ABC$  necessarily a right triangle?

**Solution for Problem 6.7:** If  $a^2 + b^2 = c^2$ , we do know that there is some triangle with side lengths  $a$ ,  $b$ ,  $c$  that is a right triangle. Specifically, consider  $\triangle DEF$  with  $\angle F = 90^\circ$ ,  $EF = a$ , and  $DF = b$ . Then, the Pythagorean Theorem tells us that  $DE^2 = EF^2 + DF^2 = a^2 + b^2$ . Since we know that  $a^2 + b^2 = c^2$ , we have  $DE = c$ . Therefore,  $\triangle DEF$  is a right triangle with sides of length  $a$ ,  $b$ , and  $c$ . Since  $\triangle ABC$  has the same side lengths as  $\triangle DEF$ , we have  $\triangle ABC \cong \triangle DEF$  by SSS. Therefore,  $\triangle ABC$  must also be a right triangle.  $\square$

**Important:** If the sides of a triangle satisfy the Pythagorean Theorem, then the triangle must be a right triangle. Any time you see triangle sides that satisfy the Pythagorean Theorem, you should mark the angle opposite the longest side as a right angle.

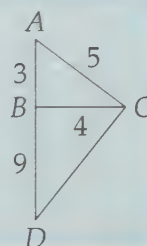


**Sidenote:** The ancient Egyptians used this method of determining a triangle is right in construction. They used loops of rope consisting of 12 rope segments of equal length. When pulled taut to form a triangle with sides equal to 3, 4, and 5 of these segments, they had a right angle opposite the 5-segment side! They could then use this to make sure angles on buildings or bricks, etc., were right angles.



Let's put this principle to work in a simple problem.

**Problem 6.8:** The side lengths are as marked in the diagram. Find  $CD$ .



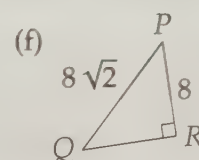
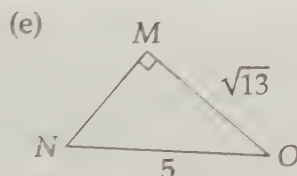
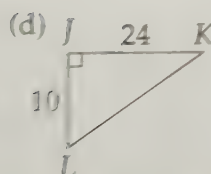
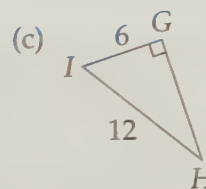
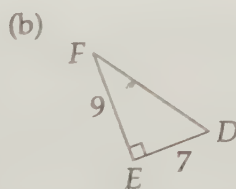
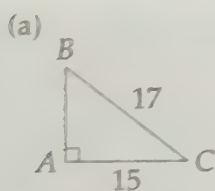
*Solution for Problem 6.8:* Since  $AB^2 + BC^2 = AC^2$ , we know that  $\triangle ABC$  is a right triangle with right angle at  $B$ . Therefore,  $\triangle CBD$  is a right triangle with hypotenuse  $\overline{CD}$ , so we can use the Pythagorean Theorem:

$$CD^2 = BC^2 + BD^2 = 16 + 81 = 97,$$

so  $CD = \sqrt{97}$ .  $\square$

### Exercises

6.1.1 Find the missing side length in each of the diagrams below:





**6.1.2** Prove that the two non-right angles of a right triangle are complementary.

**6.1.3** One leg of a right triangle is 3 cm more than 3 times the other leg, and the hypotenuse is 1 cm longer than the longest leg. Find the area of the triangle. **Hints:** 276

**6.1.4** Prove that in a right triangle, the hypotenuse is the longest side.

**6.1.5** Which of the following groups of three numbers can be the side lengths of a right triangle?

(a) 6, 8, 10.

(d)  $5/8, 3/2, 13/8$ .

(b) 4, 5, 6.

(e)  $2\sqrt{2}, 3\sqrt{2}, 5$ .

(c)  $9, 3\sqrt{3}, 6\sqrt{3}$ .

(f) 1.2, 3.5, 3.7.

**6.1.6** Find the area of a triangle with sides of length  $\sqrt{6}$ ,  $\sqrt{7}$ , and  $\sqrt{13}$ .

**6.1.7** Find the area of an isosceles triangle with two legs of length 8 and base of length 6. **Hints:** 61

**6.1.8** In this problem, we show that if  $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ , then  $A$  and  $B$  are the same distance from  $\overleftrightarrow{CD}$ .

(a) Let  $X$  and  $Y$  be on  $\overleftrightarrow{CD}$  such that  $\overline{AX} \perp \overleftrightarrow{CD}$  and  $\overline{BY} \perp \overleftrightarrow{CD}$ . Show that  $\angle XAB = 90^\circ$ .

(b) Prove that  $\triangle AXB \cong \triangle YBX$ , and use this to prove that  $A$  and  $B$  are equidistant from  $\overleftrightarrow{CD}$ .

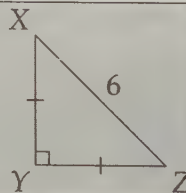
**6.1.9** Initially, a 25-foot ladder rests against a vertical wall such that the top of the ladder is 24 feet from the ground. Then, Nathan moves the base of the ladder farther out from the wall so that the top of the ladder slides down until resting against the wall at a point 20 feet above the ground. Given that the wall is perpendicular to the ground, how far did Nathan move the base of the ladder?

**6.1.10★** A triangle has sides measuring 13 cm, 13 cm, and 10 cm. A second triangle is drawn with sides measuring 13 cm, 13 cm and  $x$  cm, where  $x$  is a whole number other than 10. If the two triangles have equal areas, what is the value of  $x$ ? (Source: MATHCOUNTS) **Hints:** 161, 326

## 6.2 Two Special Right Triangles

### Problems

**Problem 6.9:**  $\triangle XYZ$  is an isosceles right triangle as shown with  $\angle Y = 90^\circ$ . Given that  $XZ = 6$ , find  $\angle X$  and  $XY$ .



**Problem 6.10:**

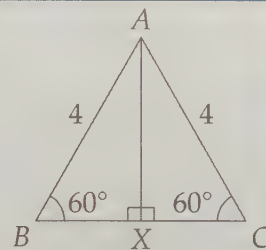
(a) If a right triangle is isosceles, must it be the two legs that are the equal sides?

(b) Show that in any isosceles right triangle, the length of the hypotenuse is  $\sqrt{2}$  times the length of a leg.



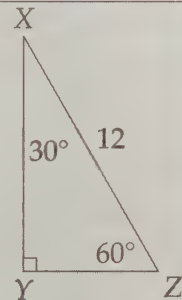
**Problem 6.11:** In this problem we will derive a formula for the area of an equilateral triangle. We'll start with  $\triangle ABC$ , an equilateral triangle with side length 4.

- Let  $\overline{AX}$  be an altitude of  $\triangle ABC$  as shown. Find  $BX$ .
- Find  $AX$ .
- Find the area of  $\triangle ABC$ .
- Let  $\triangle PQR$  be an equilateral triangle with side length  $s$ . Find  $[PQR]$  in terms of  $s$ .

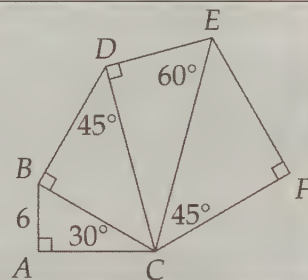


**Problem 6.12:** Our goal in this problem is to find  $YZ$  and  $YX$  in the triangle shown.

- Suppose we extend  $\overline{YZ}$  past  $Y$  to a point  $D$  such that  $YD = YZ$ . How are triangles  $\triangle XYZ$  and  $\triangle XYD$  related?
- What kind of triangle is  $\triangle XZD$ ?
- Find  $YZ$  and  $YX$ .
- How are the sides of a triangle that have angle measures  $30^\circ$ ,  $60^\circ$ , and  $90^\circ$  related?
- Extra challenge: Suppose  $\triangle PQR$  has sides 3,  $3\sqrt{3}$ , and 6. Can we deduce that the angles of  $\triangle PQR$  are  $30^\circ$ ,  $60^\circ$ , and  $90^\circ$ ?

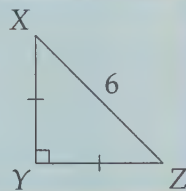


**Problem 6.13:** Find all the missing sides in the diagram.



We know a special side relationship when a triangle is isosceles, and another special relationship when a triangle is a right triangle. What if the triangle is both?

**Problem 6.9:**  $\triangle XYZ$  is an isosceles right triangle with  $\angle Y = 90^\circ$ . Given that  $XZ = 6$ , find  $\angle X$  and  $XY$ .



**Solution for Problem 6.9:** Since the triangle is isosceles, the angles opposite the equal sides are equal. Therefore,  $\angle X = \angle Z$ . Since  $\angle Y$  is a right angle,  $\angle X$  and  $\angle Z$  must add to  $90^\circ$ . So,  $\angle X = \angle Z = 90^\circ/2 = 45^\circ$ .

To find  $XY$ , we use the Pythagorean Theorem:  $XY^2 + YZ^2 = XZ^2$ . Since  $YZ = XY$  and  $XZ = 6$ , we have  $2(XY^2) = 36$ , so  $XY = \sqrt{18} = 3\sqrt{2}$ .  $\square$

This problem suggests we might be able to find a general relationship among the sides in an isosceles right triangle.

**Problem 6.10:**

- If a right triangle is isosceles, must the two sides that are equal be the legs of the triangle?
- Show that in any isosceles right triangle, the length of the hypotenuse is  $\sqrt{2}$  times the length of a leg.

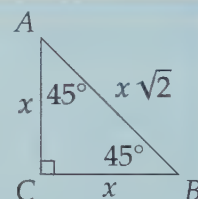
*Solution for Problem 6.10:*

- The Pythagorean Theorem quickly shows us that it must be the two legs that are equal in an isosceles right triangle. Suppose our triangle is  $\triangle ABC$  with right angle at  $\angle C$ . The Pythagorean Theorem gives us  $AC^2 + BC^2 = AB^2$ . So, if  $AB$  equals  $AC$  or  $BC$ , then the other leg must be 0, which is impossible. Therefore, in any isosceles right triangle, it is the two legs that are equal.
- If legs  $AC$  and  $BC$  each equal  $x$ , then we can find hypotenuse  $AB$  with the Pythagorean Theorem:  $AB^2 = x^2 + x^2 = 2x^2$ . Therefore,  $AB = x\sqrt{2}$ , so the hypotenuse must be  $\sqrt{2}$  times a leg.

□

**Important:**

In any isosceles right triangle, the legs are equal and the hypotenuse is  $\sqrt{2}$  times the leg. Because the angles of an isosceles right triangle are always  $45^\circ$ ,  $45^\circ$ , and  $90^\circ$ , an isosceles right triangle is sometimes called a **45-45-90 triangle**. If we know the length of any side of such a triangle, we can quickly find the others using the side relationships shown in the figure at the right.

**Sidenote:**

Due to the theorem that bears his name, **Pythagoras** may be an even more well-known name to the general public than the widely acknowledged 'father' of geometry, Euclid. Pythagoras lived during the 6th and 5th centuries BC (nearly 200 years before Euclid), and was the founder of a secret society known, unsurprisingly, as the Pythagoreans. It is unclear whether Pythagoras was the first to prove the Pythagorean Theorem, but he and his organization were definitely responsible for a number of other discoveries, including one that dramatically shook their understanding of the world.

The Pythagoreans had a deeply held belief that every phenomenon in nature could be described by whole numbers. However, their study of isosceles right triangles led them to the startling conclusion that the ratio of the length of the hypotenuse of an isosceles right triangle to the length of a leg of the triangle cannot be expressed as a ratio of two integers. As we have seen in this section, this ratio equals  $\sqrt{2}$ .

*Continued on the next page. . .*

**Sidenote:** . . . continued from the previous page



To show that  $\sqrt{2}$  cannot be written as a ratio of two integers, we use **proof by contradiction**. This means we assume the opposite of what we want to prove, and show that this assumption leads to an impossible conclusion. Here's what the proof looks like:

We assume that  $\sqrt{2} = p/q$ , where  $p$  and  $q$  are integers and  $p/q$  is in lowest terms. Squaring  $\sqrt{2} = p/q$  gives  $2 = p^2/q^2$ , so we have  $p^2 = 2q^2$ . Therefore,  $p$  must be even, so we can write  $p = 2m$  for some integer  $m$ . This gives  $(2m)^2 = 2q^2$ , so  $2m^2 = q^2$ . Therefore,  $q$  is even, too. But  $p$  and  $q$  can't both be even if  $p/q$  is in lowest terms!

Since our assumption that  $\sqrt{2}$  can be written as a ratio of integers in lowest terms leads to the impossible conclusion that the ratio is also *not* in lowest terms, we have found our **contradiction** that forces us to conclude that our assumption is wrong. Therefore,  $\sqrt{2}$  cannot be written as a ratio of integers.

We call a number that can be written as a ratio of integers a **rational number**. A number like  $\sqrt{2}$  that cannot be so written is called an **irrational number**. Some sources claim that the Pythagorean responsible for proving the existence of irrational numbers, **Hippasus**, was rewarded for his insight by being murdered. Mathematical discoveries are considerably more well-rewarded now! (See page 216.)

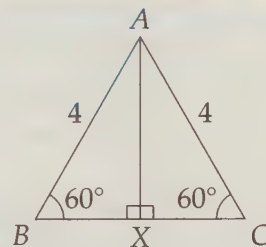
**Problem 6.11:** Find the area of an equilateral triangle with side length 4, then find a formula for the area of an equilateral triangle with side length  $s$ .

*Solution for Problem 6.11:* To find the area, we will need to find the length of an altitude. Therefore, we draw altitude  $\overline{AX}$  as shown. Since  $\angle B = \angle C$ ,  $\angle AXB = \angle AXC$ , and  $AB = AC$ , we have  $\triangle AXB \cong \triangle AXC$ . Therefore,  $BX = CX = BC/2 = 2$ . We can now use the Pythagorean Theorem to find  $AX$ . Since  $AX^2 + BX^2 = AB^2$ , we have:

$$AX = \sqrt{AB^2 - BX^2} = \sqrt{16 - 4} = \sqrt{12} = 2\sqrt{3}.$$

The area of  $\triangle ABC$  is

$$[ABC] = \frac{(BC)(AX)}{2} = \frac{(4)(2\sqrt{3})}{2} = 4\sqrt{3}.$$



We can use the exact same procedure to derive a formula for the area of an equilateral triangle with side length  $s$ . If we let the sides of  $\triangle ABC$  each be  $s$  and draw altitude  $\overline{AX}$  as before, we have  $BX = CX = s/2$ . The Pythagorean Theorem then gives

$$AX = \sqrt{AB^2 - BX^2} = \sqrt{s^2 - \frac{s^2}{4}} = \sqrt{\frac{3s^2}{4}} = \frac{s\sqrt{3}}{2}.$$

Now we can find our area:

$$[ABC] = \frac{(BC)(AX)}{2} = \frac{s(s\sqrt{3}/2)}{2} = \frac{s^2\sqrt{3}}{4}.$$

□



**Important:**


The area of an equilateral triangle with side length  $s$  is  $s^2 \sqrt{3}/4$ . If you understand how we found this formula, you shouldn't need to memorize it!

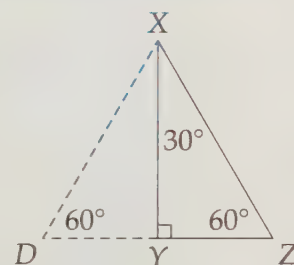
Our derivation of the formula for the area of an equilateral triangle suggests a notable relationship among the sides of a triangle with angles  $30^\circ$ ,  $60^\circ$ , and  $90^\circ$ .

**Problem 6.12:**

- How are the sides of a triangle that have angle measures  $30^\circ$ ,  $60^\circ$ , and  $90^\circ$  related?
- Suppose  $\triangle PQR$  has sides 3,  $3\sqrt{3}$ , and 6. Can we deduce that the angles of  $\triangle PQR$  are  $30^\circ$ ,  $60^\circ$ , and  $90^\circ$ ?

*Solution for Problem 6.12:*

- We found the area of an equilateral triangle by cutting the triangle into two right triangles that have acute angles of  $30^\circ$  and  $60^\circ$ . Here we go backwards – we build an equilateral triangle with two of these right triangles. In the diagram to the right, congruent right triangles  $\triangle XYZ$  and  $\triangle XYD$  together form equilateral triangle  $XDZ$ .

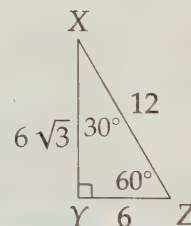


Suppose the length of  $\overline{XZ}$  is  $s$ . Since  $ZD = XZ = s$  and  $DY = YZ$ , we have  $YZ = s/2$ . As we saw in Problem 6.11, we can use the Pythagorean Theorem on  $\triangle XYZ$  to find that  $XY = s\sqrt{3}/2$ . Therefore, the sides in any triangle with angles  $30^\circ$ ,  $60^\circ$ , and  $90^\circ$  come in the ratio:

$$\text{Leg opposite } 30^\circ : \text{Leg opposite } 60^\circ : \text{Hypotenuse} = 1 : \sqrt{3} : 2.$$

In other words, the hypotenuse is double the leg opposite the  $30^\circ$  angle, and the leg opposite the  $60^\circ$  angle is  $\sqrt{3}$  times the other leg.

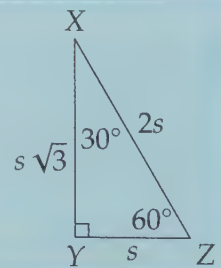
For example, if the hypotenuse is 12, then the leg opposite the  $30^\circ$  angle is  $12/2 = 6$ , and the leg opposite the  $60^\circ$  angle is  $6\sqrt{3}$  as shown in the figure to the right.



We have just discovered the relationship among the sides of another common type of right triangle.

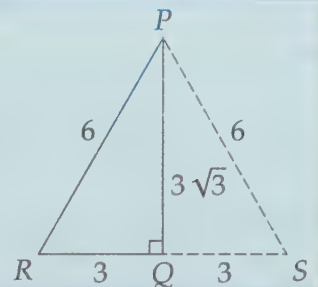
**Important:**


In a right triangle with acute angles of  $30^\circ$  and  $60^\circ$ , the sides are in the ratio  $1 : \sqrt{3} : 2$  as shown to the right. Such a triangle is often called a **30-60-90 triangle**.



- Having found this relationship among the sides given the angles, we wonder if all triangles that have sides in these ratios are 30-60-90 triangles. We'll consider one such triangle:  $\triangle PQR$  with sides 3,  $3\sqrt{3}$ , and 6. What's wrong with this 'proof' that  $\triangle PQR$  is a 30-60-90 triangle:

**Bogus Solution:** We extend  $\overline{RQ}$  to  $S$  as shown such that  $QS = QR = 3$ . Since  $QS = QR$  and  $PQ = PQ$ , we have  $\triangle PQR \cong \triangle PQS$  by SAS Congruence. Therefore,  $PS = PR = 6$ . Clearly,  $RS = 6$  also, so  $\triangle PQS$  is equilateral. So,  $\angle R = 60^\circ$  and  $\angle RPQ = 90^\circ - \angle R = 30^\circ$ , and  $\triangle PQR$  is 30-60-90.



Unfortunately, we have a big problem here: 'Since  $QS = QR$  and  $PQ = PQ$ , we have  $\triangle PQR \cong \triangle PQS$  by SAS Congruence.' This statement assumes that  $\angle PQR = \angle PQS$ , which in turn assumes that  $\angle PQR$  is a right angle. But we can't assume that, because that's what we're trying to prove!

There are a couple ways we can show that  $\triangle PQR$  is a 30-60-90 triangle.

**Solution 1:** Patch the hole in the Bogus Solution. We show that  $\triangle PQR$  is a right triangle by showing that its sides satisfy the Pythagorean Theorem:

$$3^2 + (3\sqrt{3})^2 = 9 + 27 = 36 = 6^2.$$

Therefore,  $\triangle PQR$  is a right triangle with right angle opposite the side with length 6. From here, we can use the Bogus Solution to complete our proof that  $\triangle PQR$  is a 30-60-90 triangle.

**Solution 2:** We can use triangle congruence another way to write an even simpler proof! We compare our  $\triangle PQR$  to 30-60-90 triangle  $\triangle ABC$ , which has hypotenuse 6. Because  $\triangle ABC$  is 30-60-90, we know that its legs have lengths 3 and  $3\sqrt{3}$ . Therefore,  $\triangle ABC \cong \triangle PQR$  by SSS Congruence, so the angles of  $\triangle PQR$  are the same as those of  $\triangle ABC$ . In other words,  $\triangle PQR$  is a 30-60-90 triangle.

**Important:** If the side lengths of a triangle are in the ratio  $1 : \sqrt{3} : 2$ , then the triangle is a 30-60-90 triangle, with the right angle opposite the longest side and the  $30^\circ$  angle opposite the shortest side.

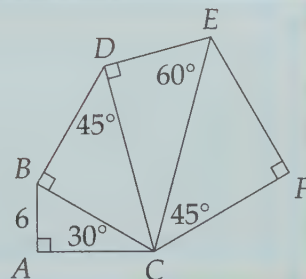


You've seen two proofs for one specific triangle above; you'll have a chance to write a proof for all such triangles in the Exercises. (You knew that was coming!)

□

Let's use our knowledge of 45-45-90 and 30-60-90 triangles to find some side lengths in a problem.

**Problem 6.13:** Find all the missing sides in the diagram.



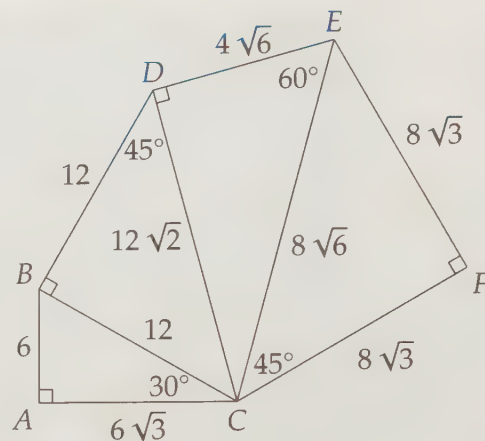
**Solution for Problem 6.13:** Because  $\triangle ABC$  is a 30-60-90 triangle and  $\overline{AB}$  is opposite the  $30^\circ$  angle, we have  $AC = AB\sqrt{3} = 6\sqrt{3}$  and  $BC = 2(AB) = 12$ .

Because  $\triangle BDC$  is a 45-45-90 triangle, we have  $BD = BC = 12$  and  $CD = BC\sqrt{2} = 12\sqrt{2}$ .

Because  $\triangle EDC$  is a 30-60-90 triangle and  $\overline{CD}$  is the longer leg, we have  $CD = DE\sqrt{3}$ . Since  $CD = 12\sqrt{2}$ , we have

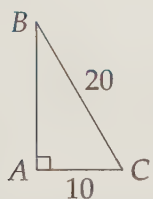
$$DE = \frac{CD}{\sqrt{3}} = \frac{12\sqrt{2}}{\sqrt{3}} = \frac{12\sqrt{2}}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{12\sqrt{6}}{3} = 4\sqrt{6}.$$

We also have  $EC = 2ED = 8\sqrt{6}$ . Finally,  $\triangle ECF$  is a 45-45-90 triangle with hypotenuse  $EC = 8\sqrt{6}$ . Therefore, the legs each have length  $EC/\sqrt{2} = 8\sqrt{3}$ . All our lengths are shown in the completed figure at right.  $\square$

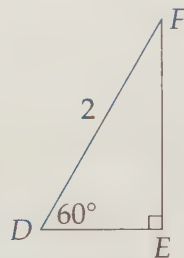


### Exercises

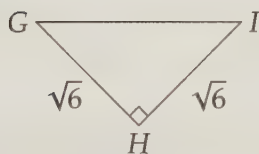
#### 6.2.1



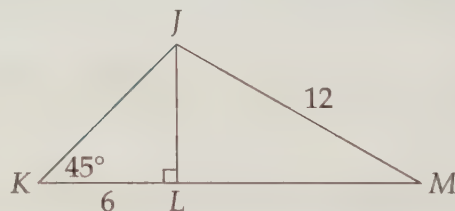
(a) Find  $AB$ ,  $\angle B$ , and  $\angle C$ .



(b) Find  $DE$ ,  $EF$ , and  $\angle F$ .



(c) Find  $GI$ ,  $\angle G$ , and  $\angle I$ .

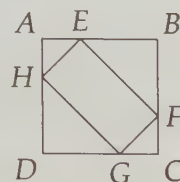


(d) Find  $JK$ ,  $JL$ ,  $LM$ ,  $\angle LJM$ , and  $\angle M$ .

**6.2.2** Prove that any triangle whose sides have lengths in the ratio  $1 : \sqrt{3} : 2$  is a right triangle. **Hints:** 413

**6.2.3** Let  $ABCD$  be a square of side length 3, and let  $E$ ,  $F$ ,  $G$ , and  $H$  be points on sides  $\overline{AB}$ ,  $\overline{BC}$ ,  $\overline{CD}$ , and  $\overline{DA}$ , respectively, such that  $AE = CF = CG = AH = 1$ .

- Prove that  $EFGH$  is a rectangle by showing that all of its angles are right angles.
- Find the area of  $EFGH$ .



**6.2.4** Find the area of  $\triangle PQR$  given that  $PQ = QR = 8$  and  $\angle PQR = 120^\circ$ . **Hints:** 312

**6.2.5★** In  $\triangle ABC$ ,  $AB = 20$ ,  $\angle A = 30^\circ$ , and  $\angle C = 45^\circ$ . Find  $BC$ . (Source: MATHCOUNTS) **Hints:** 469



## 6.3 Pythagorean Triples

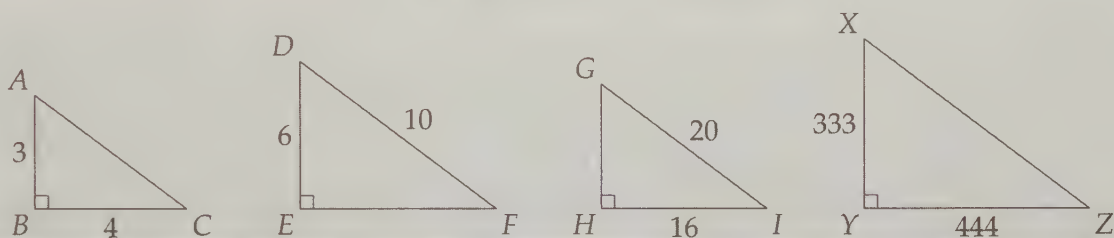
A **Pythagorean triple** is a set of three integers that satisfy the Pythagorean Theorem. For example,  $\{3, 4, 5\}$  is a Pythagorean triple because  $3^2 + 4^2 = 5^2$ . However,  $\{1, 2, 3\}$  is not a Pythagorean triple since there's no way to add two of  $1^2$ ,  $2^2$ , and  $3^2$  to get the third.

Before proceeding with the problems, play around a little and try to find some Pythagorean triples on your own. Can you find any patterns in your triples?

### Problems

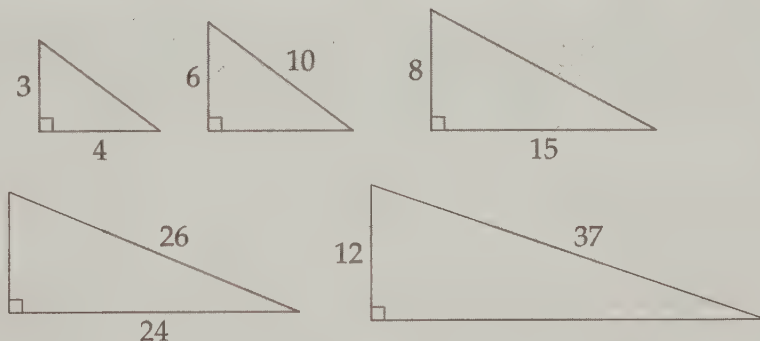
#### Problem 6.14:

- (a) Find the missing side length in each right triangle shown below.
- (b) Given that  $\{a, b, c\}$  is a Pythagorean triple, is  $\{2a, 2b, 2c\}$  a Pythagorean triple? How about  $\{na, nb, nc\}$ , where  $n$  is a positive integer?



#### Problem 6.15:

- (a) Find the last side in each right triangle below. Do you see a pattern?

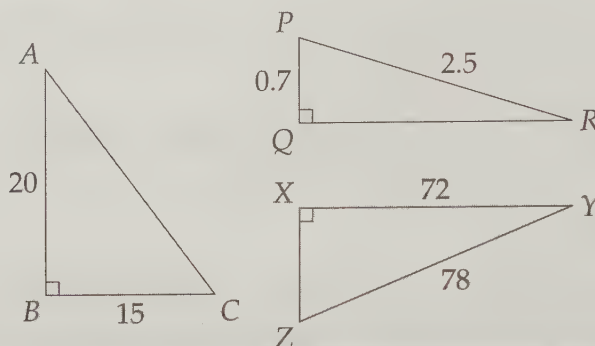


- (b) Complete the table below.

| $n$ | $2n$ | $n^2 - 1$ | $n^2 + 1$ |
|-----|------|-----------|-----------|
| 2   | 4    | 3         | 5         |
| 3   |      |           |           |
| 4   |      |           |           |
| 5   |      |           |           |
| 6   |      |           |           |

- (c) Is a triangle with sides  $2n$ ,  $n^2 - 1$ , and  $n^2 + 1$ , where  $n > 1$ , always a right triangle?

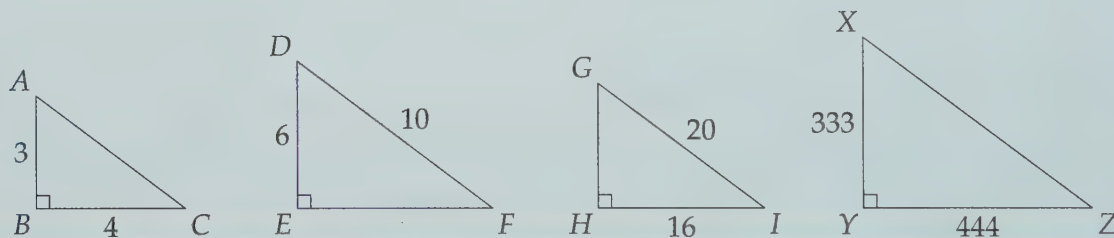
**Problem 6.16:** Find the missing side in each of the triangles below *without using the Pythagorean Theorem directly*. Just use your knowledge of Pythagorean triples!



We begin our discussion of Pythagorean triples with the most common triple.

**Problem 6.14:**

- Find the missing side length in each right triangle shown below.
- Given that  $\{a, b, c\}$  is a Pythagorean triple, is  $\{2a, 2b, 2c\}$  also a Pythagorean triple? How about  $\{na, nb, nc\}$ , where  $n$  is a positive integer?



*Solution for Problem 6.14:*

- We can apply the Pythagorean Theorem to find

$$\begin{aligned} AC &= \sqrt{AB^2 + BC^2} = \sqrt{9 + 16} = 5 \\ EF &= \sqrt{DF^2 - DE^2} = \sqrt{64} = 8 \\ GH &= \sqrt{GI^2 - HI^2} = \sqrt{144} = 12 \\ XZ &= \sqrt{XY^2 + YZ^2} = \sqrt{\text{some big numbers}} = 555 \end{aligned}$$

OK, we got a little lazy on that last one. We noted in the first three that the sides are always in the ratio  $3 : 4 : 5$ , so we took a stab at 555 for  $XZ$ . We're right, of course, but why?

There are a couple ways we can prove that we are right. First, we could note that  $\triangle XYZ \sim \triangle ABC$  due to SAS Similarity. Second, we could use algebra, as we'll see in the second part of the problem.

**Extra!** Mathematics consists of proving the most obvious thing in the least obvious way.

—George Polya

- (b) We are given that  $\{a, b, c\}$  is a Pythagorean triple, so  $a^2 + b^2 = c^2$ . To test  $\{2a, 2b, 2c\}$ , we consider the sum of the squares of the first two terms:

$$\begin{aligned}(2a)^2 + (2b)^2 &= 4a^2 + 4b^2 \\ &= 4(a^2 + b^2) \\ &= 4c^2 \text{ (because } a^2 + b^2 = c^2\text{)} \\ &= (2c)^2.\end{aligned}$$

Therefore,  $\{2a, 2b, 2c\}$  is a Pythagorean triple. There's nothing special about 2; we can run through this with any number:

$$(na)^2 + (nb)^2 = n^2a^2 + n^2b^2 = n^2(a^2 + b^2) = n^2c^2 = (nc)^2.$$

So,  $\{na, nb, nc\}$  is also a Pythagorean triple. Notice also that  $n$  doesn't have to be an integer to produce side lengths of a right triangle! For example, since a triangle with sides  $\{3, 4, 5\}$  is a right triangle, a triangle with sides  $\{3(1/7), 4(1/7), 5(1/7)\} = \{3/7, 4/7, 5/7\}$  is a right triangle.

You will often see triangles with sides in this ratio referred to as 3-4-5 triangles. Triangles whose sides are in the ratios of other Pythagorean triples are sometimes similarly identified, such as 5-12-13 triangles or 7-24-25 triangles.

□

**Important:** If  $\{a, b, c\}$  are the sides of a right triangle, then so are  $\{na, nb, nc\}$  for any positive number  $n$ .



Therefore, we can generate whole groups of Pythagorean triples from a single triple:

$$\{3, 4, 5\} \rightarrow \{6, 8, 10\} \rightarrow \{9, 12, 15\} \rightarrow \cdots \rightarrow \{333, 444, 555\} \rightarrow \cdots$$

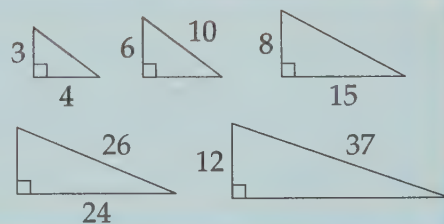
There's nothing special about integer multiples of our  $\{3, 4, 5\}$  triple. We can generate all sorts of other sets of three numbers that are the sides of a right triangle. Here are a few:

$$\{0.3, 0.4, 0.5\} \quad \{3/11, 4/11, 5/11\} \quad \{6\sqrt{5}, 8\sqrt{5}, 10\sqrt{5}\}$$

Recognizing common Pythagorean triples, or multiples of those triples, is a very useful problem solving tool. The following problem exhibits some common triples, as well as a neat pattern for generating more.

**Problem 6.15:**

- (a) Find the last side in each right triangle.  
(b) Do you see a pattern? Can you prove the pattern always works?

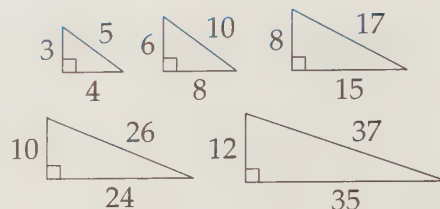


*Solution for Problem 6.15:*

- (a) We can use the Pythagorean Theorem to find the rest of the sides as shown below.



- (b) If we take the smallest even leg, halve it, then square the result, we get the number that is one more than the other leg and one less than the hypotenuse. Putting this in math terms, we let our smallest even leg have length  $2n$ . Then the other two sides are  $n^2 - 1$  and  $n^2 + 1$ . We can make a table that shows what happens for different values of the even leg.



|     | Smallest Even Leg | Other Leg | Hypotenuse |
|-----|-------------------|-----------|------------|
| $n$ | $2n$              | $n^2 - 1$ | $n^2 + 1$  |
| 2   | 4                 | 3         | 5          |
| 3   | 6                 | 8         | 10         |
| 4   | 8                 | 15        | 17         |
| 5   | 10                | 24        | 26         |
| 6   | 12                | 35        | 37         |
| 7   | 14                | 48        | 50         |

All the triangles described in the table are indeed right triangles. We can show that any triangle with sides  $2n$ ,  $n^2 - 1$ , and  $n^2 + 1$ , where  $n > 1$ , is a right triangle with some algebra:

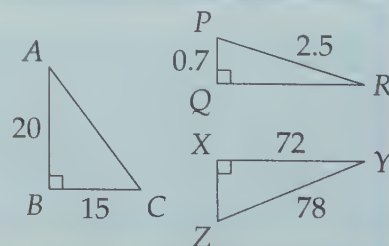
$$\begin{aligned}
 (2n)^2 + (n^2 - 1)^2 &= 4n^2 + n^4 - 2n^2 + 1 \\
 &= n^4 + 2n^2 + 1 \\
 &= (n^2 + 1)^2.
 \end{aligned}$$

□

**Sidenote:** Choose three positive integers  $k$ ,  $m$ , and  $n$ , where  $m > n$ . Compute the three numbers  $k(m^2 - n^2)$ ,  $2kmn$ , and  $k(m^2 + n^2)$ . Try it for several different values of  $k$ ,  $m$ , and  $n$ . You should find that the three numbers you generate in each case form a Pythagorean triple. See if you can figure out why this works.

Even more surprising is the fact that every single Pythagorean triple fits this mold. In other words, for every single Pythagorean triple, can we always find some  $k$ ,  $m$ , and  $n$  such that  $\{2kmn, k(m^2 - n^2), k(m^2 + n^2)\}$  generates that triple.

**Problem 6.16:** Find the missing side in each of the triangles below *without using the Pythagorean Theorem directly*. Just use your knowledge of Pythagorean triples!



**Solution for Problem 6.16:** We note that the legs of  $\triangle ABC$  are in the ratio  $15/20 = 3/4$ . Therefore, our legs are in the same ratio as the legs in a 3-4-5 triangle. We can then conclude that the hypotenuse is  $5/3$  of the smallest leg, or  $(5/3)(15) = 25$ .

We might also have found  $AC$  by noting that the legs of  $\triangle ABC$  are  $20 = 4(5)$  and  $15 = 3(5)$ , so this is a 3-4-5 triangle. Therefore, we know that  $AC = 5(5) = 25$ .

In  $\triangle PQR$ , the ratio of the given leg to the hypotenuse is  $0.7/2.5 = 7/25$ . This makes us think of the Pythagorean triple  $\{7, 24, 25\}$ . Since  $0.7 = 7(0.1)$  and  $2.5 = 25(0.1)$ , we know the other leg is  $24(0.1) = 2.4$ .

In  $\triangle XYZ$ , the ratio of the given leg to the hypotenuse is  $72/78 = 12/13$ , which reminds us of the Pythagorean triple  $\{5, 12, 13\}$ . Since our leg is  $72 = 12(6)$  and the hypotenuse is  $78 = 13(6)$ , the other leg must be  $5(6) = 30$ .

Notice our general tactic for finding the missing length in each triangle in Problem 6.16 – we take the ratio of the sides we know and then try to recognize a simple Pythagorean triple that has the numbers in our ratio.  $\square$

**WARNING!!**  Make sure you see that it's important that the 2.5 length in  $\triangle PQR$  of Problem 6.16 is the hypotenuse and not a leg. A triangle with legs 0.7 and 2.5 does not have third side 2.4. In fact, it doesn't match any simple Pythagorean triple. Therefore, you must be careful when using your knowledge of Pythagorean triples to make sure your given sides match corresponding sides (legs or hypotenuse) of the Pythagorean triple you would like to use.

**Sidenote:**  Perhaps inspired by Pythagorean Triples, the great mathematician **Pierre de Fermat** explored the equation

$$x^n + y^n = z^n$$

for values of  $n$  besides  $n = 2$  ( $n = 2$  gives the Pythagorean Theorem, of course). That there are no integers  $x$ ,  $y$  and  $z$  that satisfy this equation when  $n$  is an integer larger than 2 is known as **Fermat's Last Theorem**. Although it seems like a simple proposition, Fermat's Last Theorem evaded proof for hundreds of years until Andrew Wiles proved it in 1994. Alas, his proof is far too long to fit in the margins of this book.

### Exercises

**6.3.1** Which of the following are side lengths of a right triangle? (Try using your knowledge of Pythagorean triples!)

- (a) 300, 400, 500.
- (b) 36, 48, 60.
- (c)  $\sqrt{5}$ ,  $\sqrt{12}$ ,  $\sqrt{13}$ .
- (d) 20, 37.5, 42.5.
- (e) 1.44, 1.96, 2.4.
- (f) 15, 36, 39.

## 6.3.2

- (a) Is there a right triangle such that all three side lengths are odd integers? **Hints:** 523
- (b) Is there a right triangle such that two of the side lengths are even integers and the other is an odd integer? **Hints:** 143
- (c) Is there a right triangle in which two side lengths are simple fractions (ratios of integers, such as  $2/3$  or  $3/7$ ), and the other side length is an integer?
- (d) Is there a right triangle in which one of the side lengths is a simple fraction and the other two side lengths are integers?

**6.3.3** Susie rides her bike north 3 miles, then east 8 miles, then south 13 miles, then east 16 miles. How far is she from where she started?

**6.3.4** Use SSS Similarity to show that if  $\{a, b, c\}$  is a Pythagorean triple, then so is  $\{na, nb, nc\}$ , where  $n$  is a positive integer.

**6.3.5★** In this problem we find all Pythagorean triples in which 18 is one of the legs.

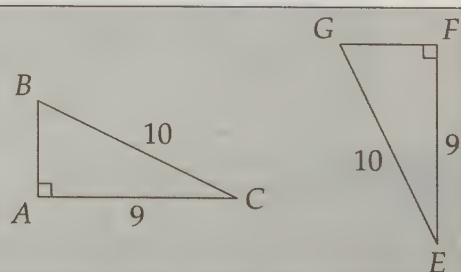
- (a) Let  $b$  and  $c$  be the other two sides, where  $c$  is the hypotenuse. Show that  $c^2 - b^2 = 18^2$ .
- (b) Factor the right side of the equation from part (a) and use the factors of  $18^2$  to solve for all possible pairs of integers of integers  $(b, c)$  that satisfy the equation.

**6.3.6★** Find the Pythagorean triple that has 97 as the length of the hypotenuse without using a computer or calculator. **Hints:** 101

## 6.4 Congruence and Similarity Revisited

## Problems

**Problem 6.17:** Are the two triangles shown congruent? Why or why not?



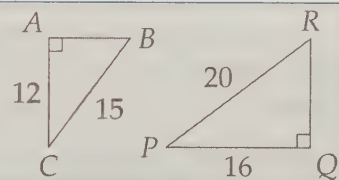
**Problem 6.18:** In this problem we will prove that if radius  $\overline{OC}$  of  $\odot O$  is perpendicular to chord  $\overline{AB}$  of  $\odot O$ , then radius  $\overline{OC}$  bisects chord  $\overline{AB}$ .

- (a) Draw a diagram to use in your proof. Include both radius  $\overline{OC}$  and chord  $\overline{AB}$ .
- (b) Find a pair of congruent triangles that you can use to complete the proof.

**Problem 6.19:** The center of a circle is 4 units away from a chord  $\overline{PQ}$  of the circle. If  $PQ = 12$ , what is the radius of the circle?

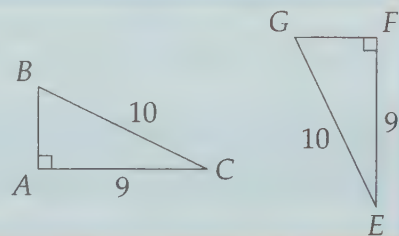


**Problem 6.20:** Are the two triangles in the diagram similar? Why or why not?



In this section we discuss a special congruence theorem and a special similarity theorem that only work for right triangles.

**Problem 6.17:** Are the two triangles in the diagram congruent? Why or why not?



*Solution for Problem 6.17:* What's wrong here:

**Bogus Solution:** Two sides and an angle of one triangle equal two sides and an angle of the other, so the triangles are congruent.



The problem here is that SSA is not in general a valid congruence theorem, as we found in Section 3.5. However, if we note that the angles that are equal are right angles...

We can use the Pythagorean Theorem to find  $AB$  and  $FG$ . In both cases, the length is  $\sqrt{100 - 81} = \sqrt{19}$ . Therefore, by SSS (or SAS), we have  $\triangle ABC \cong \triangle FGE$ .  $\square$

In the same way, any time the hypotenuse and a leg of one right triangle equal those of another, we can use the Pythagorean Theorem to show that the third sides of each triangle are equal. In other words, SSA works for right triangles. Rather than give it an unwieldy name like 'SSA, but just for right triangles', we call it:

**Important:** **Hypotenuse-Leg Congruence (HL Congruence)** states that if the hypotenuse and a leg of one right triangle equal those of another, then the triangles are congruent.



As you might guess, there's an **Leg-Leg Congruence (LL Congruence)**, but that's just SAS Congruence (make sure you see why), so there's really no need to give this a special name.

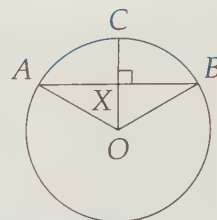
**Problem 6.18:** Prove that if radius  $\overline{OC}$  of  $\odot O$  is perpendicular to chord  $\overline{AB}$  of  $\odot O$ , then radius  $\overline{OC}$  bisects chord  $\overline{AB}$ .

**Extra!** *Common sense is the collection of prejudices acquired by age eighteen.*



—Albert Einstein

*Solution for Problem 6.18:* As usual, we start with a diagram, including both our radius and our chord. We add the radii to point  $A$  and  $B$  so we have right triangles to work with. Since  $OA = OB$  and  $OX = OX$ , we have  $\triangle OXA \cong \triangle OXB$  by HL Congruence. Therefore,  $AX = BX$ , so radius  $\overline{OC}$  bisects chord  $\overline{AB}$ .  $\square$



Problem 6.18, combined with Problem 3.4 on page 54, gives us a very useful tool:

**Important:**

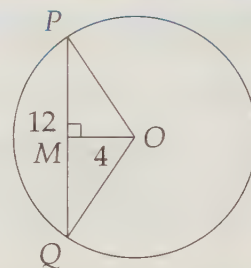


If a radius of a circle bisects a chord of the circle, it is perpendicular to the chord. Conversely, if a radius of a circle is perpendicular to the chord, it bisects the chord.

We can use this tool on a variety of circle problems. For example:

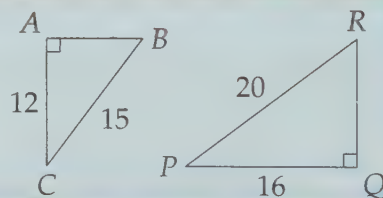
**Problem 6.19:** The center of a circle is 4 units away from a chord  $\overline{PQ}$  of the circle. If  $PQ = 12$ , what is the radius of the circle?

*Solution for Problem 6.19:* We let point  $M$  be the point on  $\overline{PQ}$  closest to the center of the circle, which we'll call  $O$ . Since  $M$  is the point on  $\overline{PQ}$  closest to  $O$ , we have  $\overline{OM} \perp \overline{PQ}$ . Because  $\overline{OM}$  is part of a radius that is perpendicular to chord  $\overline{PQ}$ ,  $\overline{OM}$  bisects chord  $\overline{PQ}$ . So, we have  $PM = MQ = 6$ . Finally, we apply the Pythagorean Theorem to  $\triangle OPM$  to find  $OP = \sqrt{PM^2 + OM^2} = \sqrt{36 + 16} = 2\sqrt{13}$ .  $\square$



Unsurprisingly, the Pythagorean Theorem can be used to give us a Hypotenuse-Leg Similarity Theorem as well.

**Problem 6.20:** Are the two triangles in the diagram similar? Why or why not?



*Solution for Problem 6.20:* What's wrong with this:

**Bogus Solution:**



Since  $20/15 = 16/12$ , we have two pairs of sides in a common ratio and  $\angle A = \angle Q$ , so by SAS Similarity, the triangles are similar.

SAS Similarity requires the angle to be *between* the two sides that have the constant ratios, so we can't use SAS Similarity like this.

We recognize these two triangles as 3-4-5 triangles (or we use the Pythagorean Theorem), and we find that  $AB = 9$  and  $QR = 12$ . Therefore, we have

$$\frac{AB}{RQ} = \frac{AC}{PQ} = \frac{BC}{PR}$$

so  $\triangle ABC \sim \triangle QRP$  by SSS Similarity.  $\square$

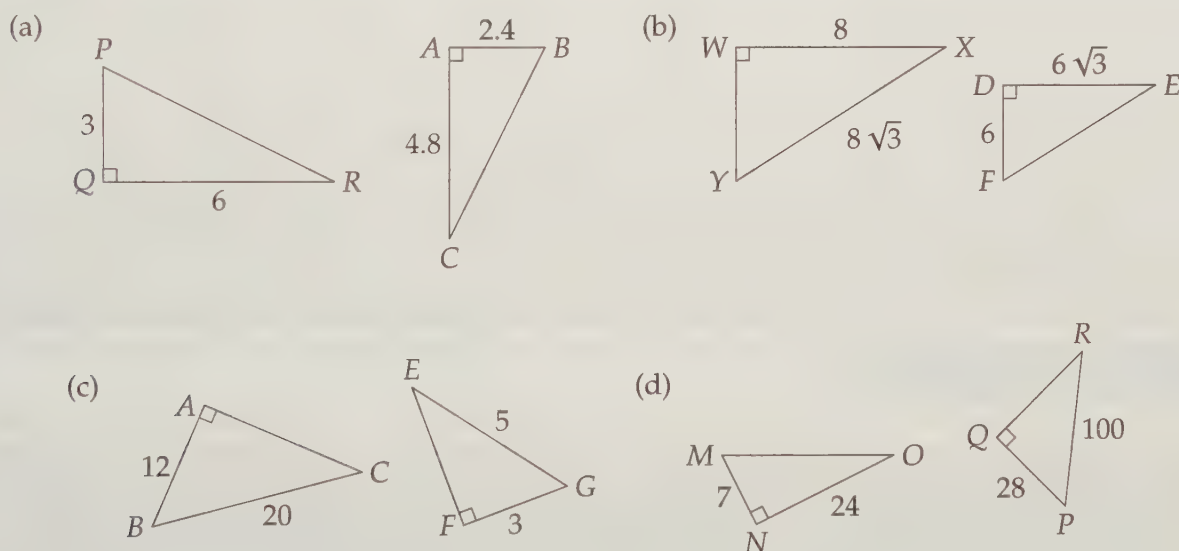
In much the same way, we can use the Pythagorean Theorem to show that if a hypotenuse and a leg of one right triangle are in the same ratio as the hypotenuse and leg of another right triangle, then the other legs are also in that ratio. So, the two triangles are similar. Therefore, we have a new similarity theorem just for right triangles.

**Important:**  **Hypotenuse-Leg Similarity (HL Similarity)** states that if the hypotenuse and a leg of one right triangle are in the same ratio as the hypotenuse and leg of another right triangle, then the triangles are similar.

As with congruence, there is a **Leg-Leg Similarity Theorem** that states that if the legs of one right triangle are in the same ratio as the legs of another right triangle, then the two triangles are similar. However, this is just the same thing as SAS Similarity. (Make sure you see why!)

### Exercises

**6.4.1** Which of the following pairs of triangles are similar? For the pairs that are similar, write both the similarity relationship and why the two triangles are similar.



**6.4.2** Points  $A$  and  $B$  are on a circle with radius 9. Given that the center of the circle is 3 units from  $\overline{AB}$ , find  $AB$ .

**6.4.3** In this problem we prove the HL Similarity Theorem we discovered in Problem 6.20. Let  $\triangle ABC$  and  $\triangle MNO$  be right triangles with right angles at  $\angle A$  and  $\angle M$ . Furthermore, suppose  $AB/BC = MN/NO$ . We wish to prove that  $\triangle ABC \sim \triangle MNO$ .

- Use the Pythagorean Theorem to find expressions for  $AC^2$  and  $MO^2$ .
- Use the given equation  $AB/BC = MN/NO$  to find an expression for  $AB$ . Write an expression equal to  $AC^2/MO^2$ .



- (c) Substitute your expression for  $AB$  into your expression for  $AC^2/MO^2$ . Prove that  $AC^2/MO^2 = BC^2/NO^2$ . **Hints:** 345
- (d) Prove that  $\triangle ABC \sim \triangle MNO$ . **Hints:** 507

**6.4.4**  $\overline{AD}$  and  $\overline{BC}$  are both perpendicular to  $\overline{AB}$  in the diagram at left below, and  $\overline{CD} \perp \overline{AC}$ . If  $AB = 4$  and  $BC = 3$ , find  $CD$ . (Source: HMMT)

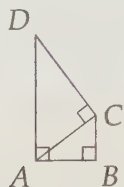


Figure 6.1: Diagram for Problem 6.4.4

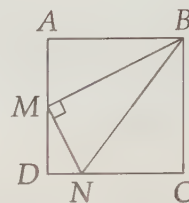


Figure 6.2: Diagram for Problem 6.4.5

**6.4.5**  $ABCD$  is a square in the diagram at right above.  $M$  is the midpoint of  $\overline{AD}$  and  $\overline{BM} \perp \overline{MN}$ .

- (a) Prove that  $\triangle MDN \sim \triangle BAM$ .
- (b)★ Prove that  $\angle ABM = \angle MBN$ . **Hints:** 189

## 6.5★ Heron's Formula

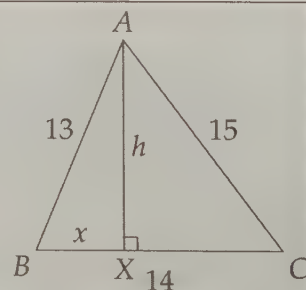
As we saw when discussing SSS Congruence, all triangles with the same three side lengths are congruent. Therefore, they all have the same area. So, we wonder, 'Since the area is fixed given the three side lengths, is there a way to figure out what the area is?'

In this section we'll answer this question with the Pythagorean Theorem and a lot of algebra.

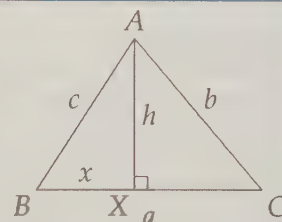
### Problems

**Problem 6.21:** In this problem, we find the area of  $\triangle ABC$ , which has sides of length 13, 14, and 15. We start by drawing altitude  $\overline{AX}$ . Let it have length  $h$  and let  $BX = x$ .

- What is  $XC$  in terms of  $x$ ?
- Apply the Pythagorean Theorem to  $\triangle ABX$  to get an equation with  $h$  and  $x$  in it.
- Apply the Pythagorean Theorem to  $\triangle ACX$  to get an equation with  $h$  and  $x$  in it.
- Subtract your equation in (c) from the one in (b) to find  $x$ . Use  $x$  to find  $h$ .
- Find the area of the triangle.
- How could you use your knowledge of Pythagorean triples to avoid all this algebra?



**Problem 6.22:** In this problem we will try to find a formula for the area of a triangle in terms of only its sides. As usual, we let  $AB = c$ ,  $BC = a$ , and  $AC = b$ . We start again with altitude  $AX = h$ , and we let  $BX = x$ . Finally, we let  $s = (a + b + c)/2$  to simplify our algebra in the problem. (Warning: There's a lot of algebra in this problem. You will use the difference of squares factorization many times:  $y^2 - z^2 = (y - z)(y + z)$ .)

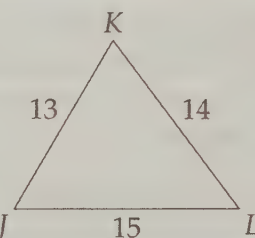
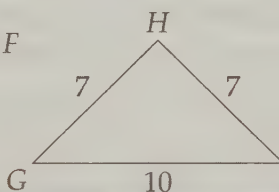
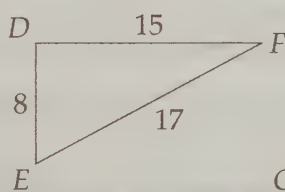
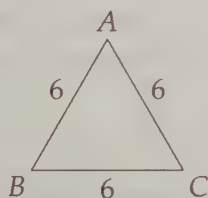


- Apply the Pythagorean Theorem to  $\triangle ABX$  to get an equation with  $a$ ,  $b$ ,  $c$ ,  $h$ , and/or  $x$  in it.
- Apply the Pythagorean Theorem to  $\triangle ACX$  to get an equation with  $a$ ,  $b$ ,  $c$ ,  $h$ , and/or  $x$  in it.
- Subtract your equation in (b) from the one in (a) to get an equation with just  $x$ ,  $a$ ,  $b$ , and  $c$ . Use this equation to show that  $x = (a^2 + c^2 - b^2)/(2a)$ .
- Substitute your expression from (c) into one of your earlier equations to show that

$$h = \frac{2\sqrt{s(s-a)(s-b)(s-c)}}{a}.$$

- Show that  $[ABC] = \sqrt{s(s-a)(s-b)(s-c)}$ .

**Problem 6.23:** Find the area of each triangle below using the formula you derived in the previous problem.



Before we tackle finding a formula for the area of a triangle in terms of its sides, we try finding the area of a specific triangle.

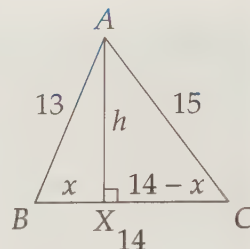
**Concept:** Often it's best to try a few examples before attempting to derive a general formula. The examples can be a good guide when you try to derive your formula.

**Problem 6.21:** Find the area of a triangle with sides of lengths 13, 14, and 15.

**Extra!** You've probably heard that music and math are related. You can thank Pythagoras for this discovery, too! Pythagoras is credited with discovering that pleasant sounds are produced by vibrating strings when the ratio of the lengths of the strings is a whole number. This discovery led to the development of the musical concept we know now as the **octave**.

*Solution for Problem 6.21:* We start by drawing altitude  $AX = h$  of  $\triangle ABC$  as shown. We let  $BX = x$ , so that  $XC = 14 - x$ . We have some right triangles, and lots of lengths, so we try the Pythagorean Theorem. Applying the Pythagorean Theorem to triangles  $\triangle AXB$  and  $\triangle AXC$  yields:

$$\begin{aligned}x^2 + h^2 &= 169 \\(14 - x)^2 + h^2 &= 225\end{aligned}$$



Seeing  $h^2$  on the left of both equations, we think to subtract the second equation from the first to eliminate  $h^2$ . We get

$$x^2 - (14 - x)^2 = 169 - 225.$$

Therefore,  $x^2 - 196 + 28x - x^2 = -56$ , so  $x = 5$ . We can substitute  $x = 5$  into either of our original equations to find  $h = 12$ . Therefore,

$$[ABC] = \frac{(BC)(AX)}{2} = \frac{(14)(12)}{2} = 84.$$

Seeing that  $x$  and  $h$  are integers, we wonder if we could have found them without the algebra. Triangles  $\triangle ABX$  and  $\triangle ACX$  share a leg and have 13 and 15 as their respective hypotenuses. This makes us think of the Pythagorean triples 5-12-13 and 9-12-15, which have the needed hypotenuses and a leg length in common. Fortunately, the two other legs, 5 and 9, add up to 14, the length of  $\overline{BC}$ . Therefore, when we glue a 5-12-13 triangle to a 9-12-15 along the leg with length 12, we get a triangle with sides 13, 14, 15.  $\square$

Unfortunately, we usually can't find Pythagorean triples that fit together nicely to make our triangle. However, we can follow our procedure from Problem 6.21 to find a formula for the area of a triangle in terms of its sides.

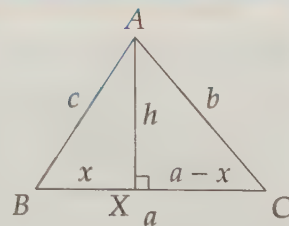
Put on your algebra hat...

**Problem 6.22:** In triangle  $ABC$ , let  $AB = c$ ,  $AC = b$ , and  $BC = a$  and let  $s = (a + b + c)/2$ . Find a formula for  $[ABC]$  in terms of  $s$ ,  $a$ ,  $b$ , and  $c$ .

*Solution for Problem 6.22:* We set up the problem in the same way as the last problem, drawing altitude  $\overline{AX}$  and letting  $AX = h$  and  $BX = x$ , so  $XC = a - x$ .

We apply the Pythagorean Theorem to  $\triangle ABX$  and  $\triangle ACX$  to get:

$$\begin{aligned}x^2 + h^2 &= c^2 \\(a - x)^2 + h^2 &= b^2\end{aligned}$$



Again, we can subtract the second equation from the first to find  $x$ :

$$x^2 - (a - x)^2 = c^2 - b^2.$$

Therefore,  $x^2 - a^2 + 2ax - x^2 = c^2 - b^2$ , so  $2ax = c^2 - b^2 + a^2$ , and we have

$$x = \frac{a^2 + c^2 - b^2}{2a}.$$



That wasn't so bad, but finding  $h$  looks a little scary. We can make the algebra somewhat nicer by solving  $x^2 + h^2 = c^2$  for  $h^2$  and factoring using the difference of squares:

$$h^2 = c^2 - x^2 = (c - x)(c + x).$$

Substituting our expression for  $x$  into this equation gives

$$\begin{aligned} h^2 &= \left( c - \frac{a^2 + c^2 - b^2}{2a} \right) \left( c + \frac{a^2 + c^2 - b^2}{2a} \right) \\ &= \left( \frac{2ac - a^2 - c^2 + b^2}{2a} \right) \left( \frac{2ac + a^2 + c^2 - b^2}{2a} \right). \end{aligned}$$

Now we recognize  $-(a - c)^2 = -a^2 + 2ac - c^2$  and  $(a + c)^2 = a^2 + 2ac + c^2$  in the numerators:

$$\begin{aligned} h^2 &= \left( \frac{b^2 - (a - c)^2}{2a} \right) \left( \frac{(a + c)^2 - b^2}{2a} \right) \\ &= \left( \frac{[b - (a - c)][b + (a - c)]}{2a} \right) \left( \frac{(a + c - b)(a + b + c)}{2a} \right) \\ &= \frac{(b + c - a)(a + b - c)(a + c - b)(a + b + c)}{4a^2}. \end{aligned}$$

(Make sure you see how we used the factorization  $y^2 - z^2 = (y - z)(y + z)$  to simplify the algebra.)

We can write  $h^2$  even more briefly with the substitution  $s = (a + b + c)/2$ . We call this  $s$  the **semiperimeter**, i.e. one-half the perimeter. From this, we have  $2s = a + b + c$ , so  $b + c - a = 2s - 2a$ . Similarly, we can substitute for all the terms in the numerator of our  $h^2$  expression:

$$h^2 = \frac{(2s - 2a)(2s - 2c)(2s - 2b)(2s)}{4a^2} = \frac{4s(s - a)(s - b)(s - c)}{a^2}.$$

Now that we have an expression for  $h$ , we can find the area:

$$[ABC] = \frac{(BC)(h)}{2} = \frac{a}{2} \sqrt{\frac{4s(s - a)(s - b)(s - c)}{a^2}} = \sqrt{s(s - a)(s - b)(s - c)}.$$

□

It's not always the nicest way to find the area of a triangle, but if we have the lengths of the three sides, we can find the area with:

**Important:** Heron's Formula tells us that if the sides of a triangle are  $a$ ,  $b$ , and  $c$ , and we let  $s = (a + b + c)/2$ , then the area of the triangle is



$$\sqrt{s(s - a)(s - b)(s - c)}.$$

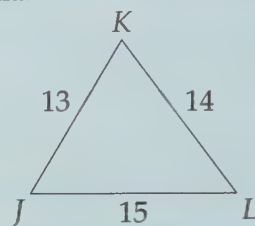
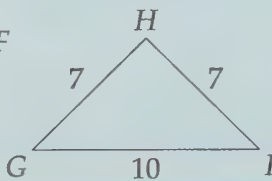
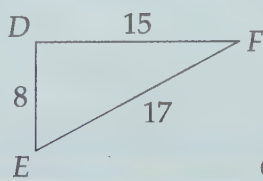
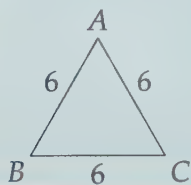
Aside from algebra, the only tool we used in deriving Heron's formula was the Pythagorean Theorem.

**Concept:** When in doubt, build right triangles and use the Pythagorean Theorem if you're trying to find a length.



Here are a few triangles on which to try Heron's formula. You should compare your answers to the areas you have found with other methods.

**Problem 6.23:** Find the area of each triangle below using Heron's formula.



*Solution for Problem 6.23:* In each case, there are other ways to find the area of the triangle. See if you can find the area using these other ways and compare your result to Heron's formula.

$$\triangle ABC: s = (6 + 6 + 6)/2 = 9, \text{ so } [ABC] = \sqrt{9(9-6)(9-6)(9-6)} = 9\sqrt{3}.$$

$$\triangle DEF: s = (8 + 15 + 17)/2 = 20, \text{ so } [DEF] = \sqrt{20(20-17)(20-15)(20-8)} = 60.$$

$$\triangle GHI: s = (7 + 7 + 10)/2 = 12, \text{ so } [GHI] = \sqrt{12(12-10)(12-7)(12-7)} = 10\sqrt{6}.$$

$$\triangle JKL: s = (13 + 14 + 15)/2 = 21, \text{ so } [JKL] = \sqrt{21(21-13)(21-14)(21-15)} = 84. \quad \square$$

Lest you start to think Heron's is always the way to go, try finding the area of a triangle with sides  $2\sqrt{3}$ , 4, and  $2\sqrt{7}$ , or a triangle with sides  $\sqrt{13}$ ,  $\sqrt{13}$ , and  $\sqrt{6}$ . Heron doesn't look so pretty in either case, but the answers can be found rather quickly with other methods. See if you can figure out how.

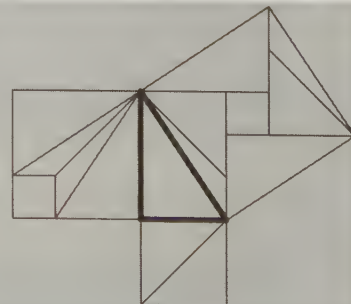
### Exercises

**6.5.1** Use Heron's Formula to find the areas of triangles with the following side lengths:

- (a) 8, 8, 8.
- (b) 4, 5, 6.
- (c) 12, 35, 37.
- (d)  $6\sqrt{2}$ ,  $7\sqrt{2}$ ,  $9\sqrt{2}$ .

**6.5.2** In  $\triangle GHI$ ,  $GH = 5$ ,  $HI = 7$ ,  $GI = 8$ . Find the height of  $\triangle GHI$  from  $G$  to  $\overline{HI}$ .

**Extra!** Chinese mathematician Liu Hui based a proof of the Pythagorean Theorem on the diagram at right in the third century A.D. (Of course, he probably didn't call the result the Pythagorean Theorem!) See if you can fill in the details!



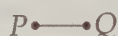
## 6.6 Construction: Perpendicular Lines

### Problems

**Problem 6.24:** Construct a line through  $A$  that is perpendicular to line  $n$ . (Hint: See Problem 3.23.)



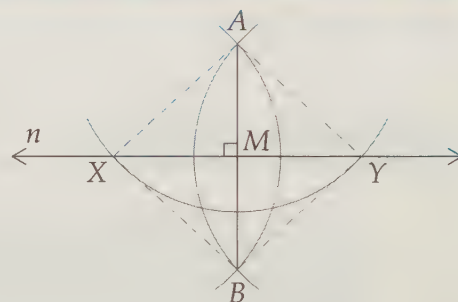
**Problem 6.25:** Suppose the shown segment has length 1. Construct a segment with length  $\sqrt{10}$ .



With all this talk about perpendicular lines and right triangles, we should wonder how to construct perpendicular lines with a ruler and compass.

**Problem 6.24:** Given line  $n$  and point  $A$ , construct a line through  $A$  that is perpendicular to  $n$ .

*Solution for Problem 6.24:* Back on page 73, we constructed a very specific perpendicular line given a segment. Maybe we can use what we learned there to make a line through  $A$  perpendicular to  $n$ . To do so, we'll have to find two points on  $n$  such that  $A$  is directly 'above' the midpoint of the segment connecting the two points. So, we draw a circle with center  $A$  that hits  $n$  in two points,  $X$  and  $Y$ . Therefore,  $AX = AY$ .



We can then mimic our construction on page 73 by drawing circles with centers  $X$  and  $Y$  and radius  $AY$ . Since  $AY = AX$ , these circles will meet at  $A$ . Let point  $B$  be the other point where these circles meet. Since  $AX = AY$ ,  $XB = YB$ , and  $AB = AB$ , we have  $\triangle AXB \cong \triangle AYB$ . Therefore,  $\angle MAX = \angle MAY$ . Together with  $AX = AY$  and  $AM = AM$ , this gives us  $\triangle MAX \cong \triangle MAY$  by SAS Congruence. Finally, we have  $\angle XMA = \angle YMA$ , so these angles are each  $90^\circ$  because together they make up a straight angle.

Therefore,  $\overleftrightarrow{AB}$  passes through  $A$  and is perpendicular to  $n$ .  $\square$

Now that we can create perpendiculars, we can use right triangles in our constructions. Let's give it a try.

**Problem 6.25:** Suppose the shown segment has length 1. Construct a segment with length  $\sqrt{10}$ .

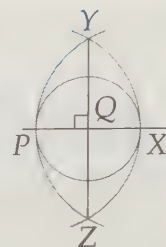


**Extra!** *The mind is not a vessel to be filled, it is a fire to be kindled.*

—Plutarch

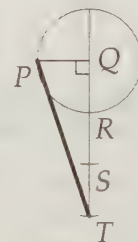


*Solution for Problem 6.25:* Seeing the square root, we think of the Pythagorean Theorem, since we've seen a lot of square root signs in our right triangle problems. Since  $1^2 + 3^2 = 10$ , we want to make a right triangle with legs of length 1 and 3. The hypotenuse of this triangle will have length  $\sqrt{10}$ . We start by extending  $\overrightarrow{PQ}$ , then constructing a line through  $Q$  perpendicular to  $\overrightarrow{PQ}$  as described in Problem 6.24. Specifically, our steps for constructing the perpendicular line are:



- Draw a circle with center  $Q$  and radius  $QP$ . Name the second point where this circle hits  $\overrightarrow{PQ}$  point  $X$ .
- Draw circles with centers  $P$  and  $X$ , both with radius  $PX$ . Name the points where these circles meet  $Y$  and  $Z$ .
- Draw  $\overleftrightarrow{YZ}$ , which is the line through  $Q$  perpendicular to  $\overrightarrow{PQ}$ . (Make sure you see why this line must go through  $Q$ .)

Once we have this perpendicular line, we create a segment of length 3 along it by drawing a circle with center  $Q$  and radius  $QP$  to get point  $R$  on the perpendicular such that  $QR = 1$ . Then we can copy  $\overline{QR}$  a couple times to get  $S$  and  $T$  such that  $RS = ST = 1$ , so  $QT = 3$ . Segment  $\overline{PT}$  is then the hypotenuse of a right triangle with legs of lengths 1 and 3, so it has length  $\sqrt{1^2 + 3^2} = \sqrt{10}$ .  $\square$



## Exercises

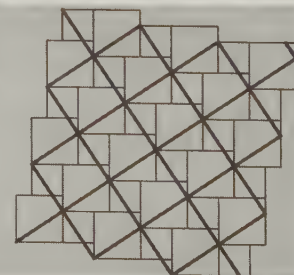
6.6.1 Construct a 45-45-90 triangle and a 30-60-90 triangle.

6.6.2 Given a segment of length 1, construct a segment of length  $\sqrt{3}$ .

## 6.7 Summary

**Definitions:** A **right triangle** is a triangle that has a right angle among its angles. The side opposite the right angle is called the **hypotenuse** and the other two sides are the **legs**.

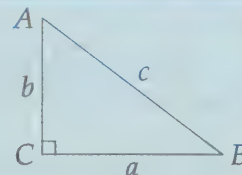
**Extra!** The dazzling tiling 'proof without words' of the Pythagorean Theorem shown at right comes from **Annairizi of Arabia** (circa 900 AD). See if you can figure out how it works! *Source: Proofs Without Words II by Roger Nelsen*



**Important:**

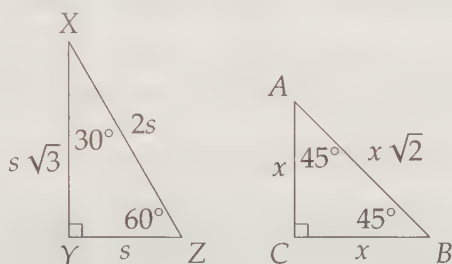
In any right triangle, the sum of the squares of the legs equals the square of the hypotenuse. Or, in the figure to the right, we have

$$a^2 + b^2 = c^2.$$



This is the famous **Pythagorean Theorem**. The Pythagorean Theorem also runs in reverse: If the sides of a triangle satisfy the Pythagorean Theorem, then the triangle must be a right triangle. The longest side of such a triangle is the hypotenuse, and the angle opposite it is a right angle.

Two special right triangles, the 45-45-90 triangle and the 30-60-90 triangle, are shown below. The relationships among the sides of each triangle are as indicated.



Conversely, if you determine that the side lengths of a triangle are in the ratio  $1 : 1 : \sqrt{2}$ , you can deduce that the triangle is a 45-45-90 triangle. If the side lengths are in the ratio  $1 : \sqrt{3} : 2$ , then the triangle is a 30-60-90 triangle with the  $30^\circ$  angle opposite the shortest side.

**Important:**

We used the Pythagorean Theorem to prove the following useful facts:

- The distance from a point to a line is the length of the segment from the point to the line that is perpendicular to the line.
- The area of an equilateral triangle with side length  $s$  is  $s^2 \sqrt{3}/4$ . If you understand how we found this formula, you shouldn't need to memorize it!
- **Heron's Formula** tells us that if the sides of a triangle are  $a$ ,  $b$ , and  $c$ , and we let  $s = (a + b + c)/2$ , then the area of the triangle is

$$\sqrt{s(s-a)(s-b)(s-c)}.$$

**Definition:** A **Pythagorean triple** is a set of three integers that satisfy the Pythagorean Theorem. Some common Pythagorean triples are:

$$\{3, 4, 5\} \quad \{5, 12, 13\} \quad \{7, 24, 25\} \quad \{8, 15, 17\} \quad \{9, 40, 41\}.$$

**Important:** If  $\{a, b, c\}$  are the sides of a right triangle, then so are  $\{na, nb, nc\}$  for any positive number  $n$ .



We also investigated two new congruence theorems and two new similarity theorems specifically for right triangles:

- **HL Congruence.** If the hypotenuse and a leg of one right triangle equal those of another, then the triangles are congruent.
- **LL Congruence.** If the legs of one right triangle equal those of another, then the triangles are congruent. (This is the same as SAS Congruence.)
- **HL Similarity.** If the hypotenuse and a leg of one right triangle are in the same ratio as the hypotenuse and leg of another right triangle, then the triangles are similar.
- **LL Similarity.** If the legs of one right triangle are in the same ratio as the legs of another right triangle, then the triangles are similar. (This is the same as SAS Similarity.)

We used HL Congruence to prove the following useful fact:

**Important:** If a radius of a circle bisects a chord of the circle, it is perpendicular to the chord. Conversely, if a radius of a circle is perpendicular to the chord, it bisects the chord.



## Problem Solving Strategies

### Concepts:



- Building right triangles and applying the Pythagorean Theorem is one of the most common ways to find lengths. This is particularly true in problems that already involve right angles.
- Often it's best to try a few examples before trying to derive a general formula – the examples can be a good guide when you try to derive your formula.

## Things To Watch Out For!

### WARNING!!



Be careful when using your knowledge of Pythagorean triples to make sure your given sides match corresponding sides (legs or hypotenuse) of the Pythagorean triple you would like to use.

**Extra!** *In mathematics you don't understand things. You just get used to them.*



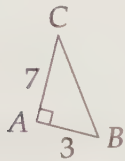
–John von Neumann



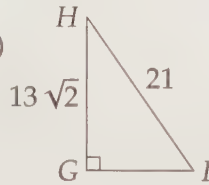
## REVIEW PROBLEMS

6.26 Find the missing side length in each of the triangles below:

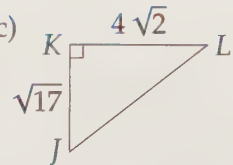
(a)



(b)

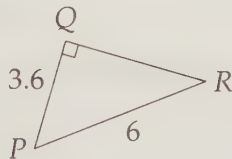


(c)

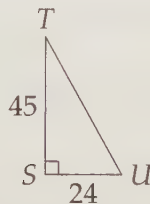


6.27 Find the missing side length in each of the triangles below by using your knowledge of Pythagorean triples:

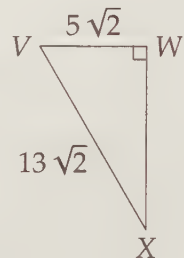
(a)



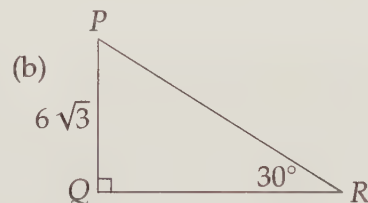
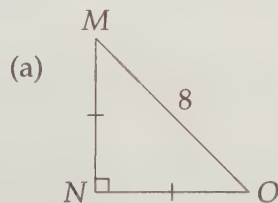
(b)



(c)



6.28 Find the missing side lengths in the triangles below:

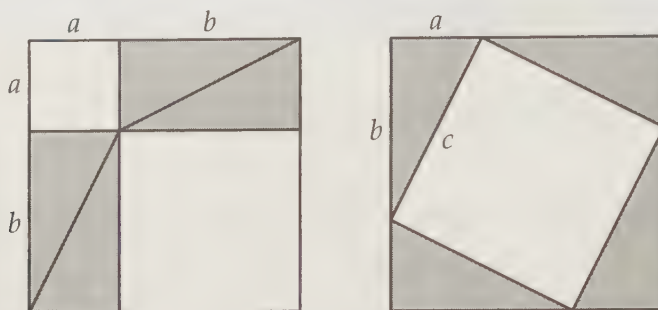


6.29 For each of the following parts, state whether or not the three numbers listed could be the side lengths of a right triangle.

- (a) 11, 16, 19.
- (b)  $\frac{1}{5}$ ,  $\frac{4}{15}$ ,  $\frac{1}{3}$ .
- (c)  $\sqrt{73}$ ,  $2\sqrt{2}$ , 9.
- (d)  $\sqrt{0.5}$ ,  $\sqrt{1.2}$ ,  $\sqrt{1.3}$ .
- (e) 0.77, 2.64, 2.75.

6.30 Given that  $AB = 3$ ,  $BC = 6$ , and  $\angle ABC = 60^\circ$ , why must  $\triangle ABC$  be a right triangle? (You cannot simply say 'It is a 30-60-90 triangle; you must prove that it is!')

6.31 In this problem, we find yet another proof of the Pythagorean Theorem.

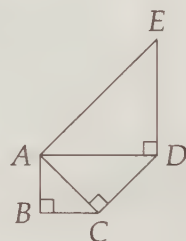


- Construct a square of side length  $a + b$ , and place four right triangles with legs of lengths  $a$  and  $b$  in it as shown in the left diagram. Let  $K$  be the area of each shaded triangle. Express the area of the whole square in terms of  $K$  and the areas of the two interior squares.
- Now place the four triangles as shown in the right diagram and let  $c$  be the length of the hypotenuse of each of the triangles. Express the area of the large square in terms of  $K$  and the area of the interior square.
- What do you conclude from parts (a) and (b)?

6.32 In the figure at right,  $AB = BC = 1$ .  $\overline{AD} \parallel \overline{BC}$  and  $\overline{CD} \parallel \overline{AE}$ . Find  $AE$ . (Source: MATHCOUNTS)

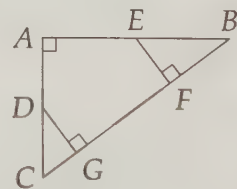
6.33 Find the perimeter of an isosceles triangle with base length 10 and area 60.

6.34 Using any method you like, find the areas of the triangles with the side lengths below:



- 6, 8, 10.
- 3, 4, 6.
- $\sqrt{6}$ ,  $\sqrt{24}$ ,  $\sqrt{30}$ .
- $1, 24/7, 25/7$ .
- 5, 6, 7.

6.35 In the figure shown,  $E$  is the midpoint of  $\overline{AB}$ ,  $D$  is the midpoint of  $\overline{AC}$ ,  $AB = 16$ , and  $AC = 12$ . (Source: MATHCOUNTS)



- Show that  $\triangle EFB \sim \triangle CAB$ .
- Find  $DG$  and  $EF$ .
- What is the area of  $AEFGD$ ?

6.36 Find the area of an equilateral triangle with a height of length 8.

6.37 In  $\triangle MNO$ ,  $MN = 13$ , and  $NO = 37$ .  $P$  is on  $\overline{MO}$  such that  $\overline{NP} \perp \overline{MO}$  and  $NP = 12$ . Find the area of  $\triangle MNO$ . (Source: MATHCOUNTS)

6.38 Let  $ABC$  be a right triangle, with  $\angle ACB = 90^\circ$ . Let  $D$  be the foot of the perpendicular from  $C$  to side  $\overline{AB}$ . Let  $x = AD$  and  $y = BD$ . Prove that  $CD = \sqrt{xy}$ .

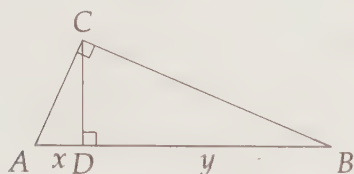


Figure 6.3: Diagram for Problem 6.38

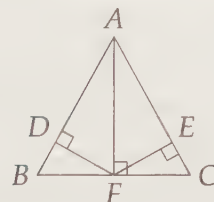
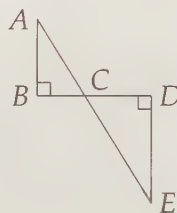


Figure 6.4: Diagram for Problem 6.39

6.39 In the diagram at right above,  $AB = AC$ ,  $\overline{AF} \perp \overline{BC}$ ,  $\overline{FD} \perp \overline{AB}$ , and  $\overline{FE} \perp \overline{AC}$ . Prove that  $FD = FE$ .

6.40 In the diagram at right,  $BD = 6$ ,  $AB = 3$ , and  $DE = 5$ . What is the length of  $\overline{AE}$ ? (Source: MATHCOUNTS)



6.41 The length of one leg of a right triangle is 22, and the other two sides also have integer lengths. Find the perimeter of the triangle.

6.42 Point  $A$  is on side  $\overline{XZ}$  of  $\triangle XYZ$  such that  $XA = XY$ . Given that  $\angle YXZ = 90^\circ$ ,  $\angle YZX = 30^\circ$ , and  $ZA = 6 - \sqrt{12}$ , find the area of  $\triangle XYZ$ .

6.43 The sides of a triangle have lengths of 15, 20, and 25. Find the length of the shortest altitude of the triangle. (Source: AMC 10)

## Challenge Problems

6.44 A right triangle has area 210 and hypotenuse 29. Find the perimeter of the triangle. **Hints:** 119, 392

6.45 What's wrong with the diagram at left below? **Hints:** 409

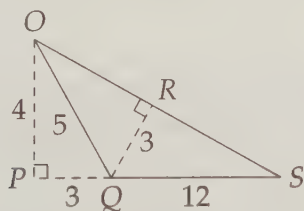


Figure 6.5: Diagram for Problem 6.45

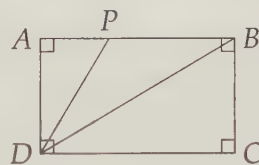


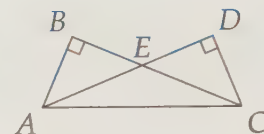
Figure 6.6: Diagram for Problem 6.46

6.46 In rectangle  $ABCD$ ,  $AD = 1$ ,  $P$  is on  $\overline{AB}$ , and  $\overline{DB}$  and  $\overline{DP}$  trisect  $\angle ADC$ . (An angle is trisected when it is divided in three equal angles.) What is the perimeter of  $\triangle BDP$ ? (Source: AMC 10) **Hints:** 421

6.47 The hypotenuse of a right triangle has length 8. The triangle's area is also 8. Find the perimeter of the triangle. **Hints:** 16



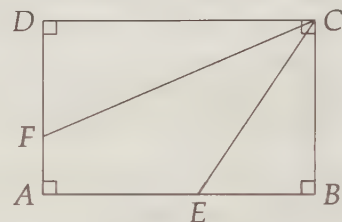
6.48 In the diagram at right  $\angle B = \angle D = 90^\circ$ ,  $AB = DC = 24$ , and  $BC = AD = 32$ . (Source: ARML)



- Prove that  $\triangle ABE \cong \triangle CDE$ .
  - Let  $M$  be the midpoint of  $\overline{AC}$ . Prove that  $\overline{EM} \perp \overline{AC}$ .
  - Find the area of  $\triangle AEC$ . **Hints:** 296
- 6.49 Triangle  $ABC$  has  $AB = 12$ ,  $BC = 16$ , and  $AC = 20$ . If  $D$  is on  $\overline{AC}$  such that  $AD = 12$ , find the area of  $\triangle ADB$ . (Source: ARML) **Hints:** 116, 424

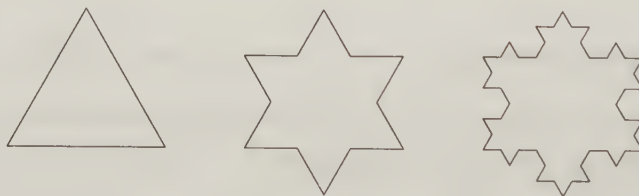
6.50 Riders on a Ferris wheel travel in a circle in a vertical plane. A particular wheel has radius 20 feet and revolves at the constant rate of one revolution per minute. How many seconds does it take a rider to travel from the bottom of the wheel to a point 10 vertical feet above the bottom? (Source: AMC 10) **Hints:** 465

6.51 Shown at right is rectangle  $ABCD$ . Angle  $C$  is trisected by  $\overline{CF}$  and  $\overline{CE}$ , where  $E$  is on  $\overline{AB}$ ,  $F$  is on  $\overline{AD}$ ,  $BE = 6$ , and  $AF = 2$ . (Source: AMC 12)



- Find  $BC$ .
  - Find  $DF$ .
  - Find  $[ABCD]$ .
- 6.52 Initially, a fifty-foot ladder rests against a wall. As I start to climb it, the ladder slides down, finally stopping such that it touches the wall at a point 8 feet below where it originally touched the wall. During the slide, the base of the ladder slid 16 feet from its original position. How far is the top of the ladder from the ground after the slide, given that the wall is perpendicular to the ground? **Hints:** 137
- 6.53 Using only a pencil and paper, find the Pythagorean triple with 73 as the length of the hypotenuse. **Hints:** 40, 423
- 6.54★ If the lengths of the three sides of a right triangle are whole numbers, how many such distinct noncongruent right triangles exist having one leg with a length of 24 units?
- 6.55 In  $\triangle RST$ ,  $RS = 13$ ,  $ST = 14$ , and  $RT = 15$ .

- Find the length of the height from  $R$  to  $\overline{ST}$ .
  - ★ Let  $M$  be the midpoint of  $\overline{ST}$ . Find  $RM$ . **Hints:** 563
- 6.56★ We begin with an equilateral triangle with side length 1. We divide each side into three segments of equal length, and add an equilateral triangle to each side using the middle third as a base. We then repeat this, to get a third figure.



If we continue this process forever, what is the area of the resulting figure? **Hints:** 13, 398

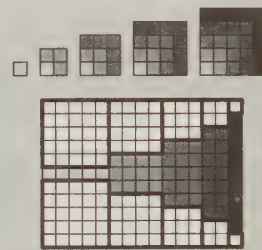
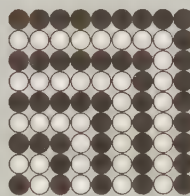
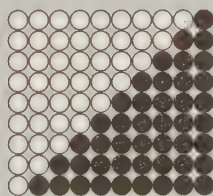
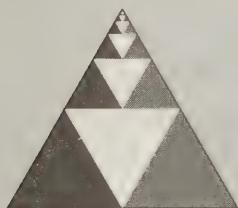
**6.57★** Let  $\triangle XOY$  be a right-angled triangle with  $\angle XOY = 90^\circ$ . Let  $M$  and  $N$  be the midpoints of legs  $\overline{OX}$  and  $\overline{OY}$ , respectively. Given that  $XN = 19$  and  $YM = 22$ , find  $XY$ . (Source: AMC 10) **Hints:** 290, 505

**6.58★** In triangle  $ABC$ ,  $AB = AC$ ,  $BC = \sqrt{3} - 1$ , and  $\angle BAC = 30^\circ$ . Find the length of  $AB$ . **Hints:** 384

**6.59★** In  $\triangle ABC$ ,  $AB = 6$  and  $BC = 8$ . Find  $AC$  in each of the following cases:

- (a)  $\angle B = 30^\circ$ . **Hints:** 57  
 (b)  $\angle B = 45^\circ$ . **Hints:** 129  
 (c)  $\angle B = 135^\circ$ . **Hints:** 215

**Extra!** The great modern mathematician Paul Erdős was fond of describing particularly beautiful proofs as being from 'the book.' For sport, many aesthetically inclined math fans find beautiful 'proofs without words' by using diagrams designed to 'show' the truth of a mathematical statement rather than 'saying' it. In our various proofs without words of the Pythagorean Theorem, we've seen proofs without words applied to the most natural area of mathematics for them, geometry. However, the diagrams below show that the tools of geometry can be used to 'prove without words' statements from other fields of mathematics, as well. Perhaps you'll agree that these proofs are indeed 'from the book.'



Match the images above with the following mathematical statements:

- $1^2 + 2^2 + 3^2 + 4^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$ .
- $\frac{1}{4} + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^3 + \left(\frac{1}{4}\right)^4 + \cdots = \frac{1}{3}$ .
- $1 + 2 + 3 + 4 + \cdots + n = \frac{n(n+1)}{2}$ .
- $1 + 3 + 5 + 7 + 9 + \cdots + (2n-1) = n^2$ .

Make sure you see how each statement is illustrated by its matching image!

Source: *Proofs Without Words* by Roger Nelsen